LINEAR ALGEBRAIC GROUPS

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Abstract. These are notes for a graduate course on linear algebraic groups taught at Zhejiang University. They were written by Zhu Ruizhi during class for which I thank him sincerely. Any comments are highly appreciated; please send them directly to mihatsch@zju.edu.cn.

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1. Introduction (9/16/2026)

Course Information

- Prerequisites: Commutative Algebra or Introduction to Algebraic Geometry
- One lecture per week, on Wednesdays 10:00 a.m. to 12:30 p.m.
- Homework graded for feedback, no effect on final grade
- Two exams:
  - 1st on November 4th, 40%.
  - 2nd on December 30th, 60%.

Date: Winter term 2026.

- Main reference: Humphrey's Linear Algebraic Groups [1]

The goal of this first lecture are to explain what the theory of linear algebraic groups is about, to introduce affine varieties, and to give the definition of a linear algebraic group.

1.1. **What is the theory about?** Mathematicians and physicist are interested in groups such as GL_n, SL_n, SO_n, or Sp_{2n}. These arise in many different contexts and with different flavors:

(1) Finite groups of Lie type $\mathrm{GL}_n(\mathbb{F}_q), \mathrm{Sp}_{2n}(\mathbb{F}_q)$,... come from a more set-theoretic, combinatorial perspective. They make up part of classification of the finite simple groups.
(2) Real or complex Lie groups $\mathrm{GL}_n(\mathbb{R}), \mathrm{SO}_n(\mathbb{C}), \mathrm{SL}_n(\mathbb{R}), \mathrm{PGL}_n(\mathbb{C})$,... arise from a more analytic perspective which is relevant in differential geometry, representation theory, etc.

Our perspective. We want to study these groups as algebraic objects. An advantage of this approach is that it makes sense over various fields. For example, we can study $\mathrm{SO}(V)/\mathbb{Q}, \mathrm{GL}_n/\mathbb{Q}_p, \mathrm{Sp}_{2n}/\mathbb{C}$ which is clearly relevant for areas such as number theory, commutative algebra, algebraic geometry, and so on.

Example 1.1. We consider $\mathrm{SL}_2/\mathbb{C}$ as algebraic or complex Lie group, and are interested in its representation theory.

By definition, a representation of SL_2 is a finite-dimensional $\mathbb{C}$-vector space V together with a homomorphism $\mathrm{SL}_2 \to \mathrm{GL}(V)$. Such representations can be completely classified:

Theorem 1.2. *Let V be the finite-dimensional representation of $\mathrm{SL}_2/\mathbb{C}$.*

(a) V is completely reducible*, meaning it is a direct sum of irreducible representations,*

$$V \cong \bigoplus_{i=1}^{r} V_i.$$

(b) Each irreducible summand V_i is isomorphic to the n-th symmetric power $\mathrm{Sym}^n(\mathbb{C}^2)$ of the standard representation, for some n.

Explanation. What is $\mathrm{Sym}^n(\mathbb{C}^2)$?

(1) $\mathrm{SL}_2 \curvearrowright \mathbb{C}^2$ is the *standard representation*

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}. \tag{1.1}$$

(2) $\mathrm{Sym}^n(\mathbb{C}^2)$ is derived from $\mathbb{C}^2$ as the space of homogeneous degree n polynomials in x and y:

$$\mathrm{Sym}^n(\mathbb{C}^2) = \{a_n x^n + a_{n-1} x^{n-1} y + \cdots + a_0 y^n \mid a_n, \ldots, a_0 \in \mathbb{C}\}.$$

SL_2 acts by the coordinate substitution given by (1.1):

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot p(x, y) = p(ax + by, cx + dy).$$

For example,

$$\begin{pmatrix} 1 & 1 \\ & 1 \end{pmatrix} \cdot x^2 = (x + y)^2 = x^2 + 2xy + y^2.$$

Theorem 1.2 is a beautiful example of a complete classification result. But such behavior is not automatic for algebraic groups, as the next example shows.

Example 1.3. Consider the *additive group* $\mathbb{G}_a = (\mathbb{C}, +)$. It has the two-dimensional representation

$$\mathbb{G}_a \longrightarrow \mathrm{GL}_2, \qquad t \longmapsto \begin{pmatrix} 1 & t \\ & 1 \end{pmatrix}.$$

- This representation is not irreducible: $\mathbb{C} \cdot \left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right) \subseteq \mathbb{C}^2$ is a $\mathbb{G}_a$-stable subspace.
- But $\mathbb{C} \cdot \left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right)$ has no invariant complement:

$$\left\langle \begin{pmatrix} a \\ b \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ & 1 \end{pmatrix} \cdot \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a+b \\ b \end{pmatrix} \right\rangle = \mathbb{C}^2 \qquad \text{if } b \neq 0.$$

Question. Which groups behave like SL_2, which like $\mathbb{G}_a$?

General theory answers this question:

- *reductive groups* behave like SL_2 (completely reducible representation theory)
- *unipotent groups* behave like $\mathbb{G}_a$ (every rep has a fixed vector like $\left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right) \in \mathbb{C}^2$ before)

Example 1.4. The groups $\mathrm{GL}_n, \mathrm{SL}_n, \mathrm{PGL}_n, \mathrm{SO}_n, \mathrm{Sp}_{2n}$ are all reductive.

Upper triangular matrices form a unipotent group, e.g.

$$\left\{ \begin{pmatrix} 1 & x & y \\ & 1 & z \\ & & 1 \end{pmatrix} \right\} \subseteq \mathrm{GL}_3.$$

Question. OK, but I've already seen all these groups before. Can you teach me some new ones?

Answer. General theory says that these examples are the building blocks for *all* LAGs!

Theorem 1.5. *Let $G/\mathbb{C}$ be a connected linear algebraic group.*

(a) There exists a unique maximal, normal, unipotent subgroup $G_u \subseteq G$ called the unipotent radical *of G. The quotient $G_{\mathrm{red}} = G/G_u$ is reductive.*

(b) The unipotent part G_u is built as a successive extension of copies of $\mathbb{G}_a$. The reductive quotient G_{red} is built out of $\mathrm{GL}_n, \mathrm{SL}_n, \mathrm{SO}_n$ and Sp_{2n}.

Example 1.6. Consider the block upper triangular subgroup

$$G := \left\{ \begin{pmatrix} x_{11} & x_{12} & y \\ x_{21} & x_{22} & z \\ 0 & 0 & x_{33} \end{pmatrix} \right\} \subseteq \mathrm{GL}_3.$$

Then

$$G_u = \left\{ \begin{pmatrix} 1 & 0 & y \\ & 1 & z \\ & & 1 \end{pmatrix} \right\} \cong \mathbb{G}_a^2, \qquad G_{\mathrm{red}} \cong \mathrm{GL}_2 \times \mathrm{GL}_1.$$

Example 1.7. The *projective linear group PGL_n* is reductive and obtained as quotient

$$\mathrm{SL}_n \xrightarrow{n:1} \mathrm{PGL}_n.$$

The *spinor group* Spin_n is reductive and obtained as a double cover

$$\mathrm{Spin}_n \xrightarrow{2:1} \mathrm{SO}_n, \qquad n \geq 3.$$

The general theory also more precisely classifies all reductive groups and their representations up to isomorphism, which we will see towards the end of the course.

To end this introduction, let us consider a different type of statement.

Question. How to work with group elements "like a professional"?

Example 1.8. Any $g \in \mathrm{GL}_n(\mathbb{C})$ can be written uniquely as a product $g = g_s \cdot g_u$ such that

- g_s is diagonalizable (also called *semi-simple*)
- g_u is unipotent (meaning $g_u - 1 \in \mathrm{M}_n(\mathbb{C})$ is nilpotent)
- $g_s \cdot g_u = g_u \cdot g_s$

Indeed, we know that g has a unique *Jordan normal form* up to permutation of blocks, and the decomposition $g = g_s \cdot g_u$ is immediate to see in JNF. For example,

$$g = \begin{pmatrix} 2 & 5 & \\ & 2 & \\ & & -1 \end{pmatrix} = \underbrace{\begin{pmatrix} 2 & & \\ & 2 & \\ & & -1 \end{pmatrix}}_{g_s} \cdot \underbrace{\begin{pmatrix} 1 & 5/2 & \\ & 1 & \\ & & -1 \end{pmatrix}}_{g_u}.$$

Here, note that g_s and g_u indeed commute as required.

Theorem 1.9. *Let G be a connected LAG over $\mathbb{C}$. Any $g \in G$ has a unique decomposition $g = g_s g_u$ into a semi-simple and unipotent part, characterized by*

$$\forall\ \rho : G \longrightarrow \mathrm{GL}_n, \qquad \rho(g) = \rho(g_s) \cdot \rho(g_u)$$

is the decomposition from Example 1.8 $\rho(g)$.

Conclusion. Summing up this lecture, we have seen that the theory of LAGs

- Classifies and describes algebraic groups such as $\mathrm{GL}_n, \mathrm{SL}_n, \mathrm{Spin}_n$,
- Classifies and describes their representations, and
- Provides structural tools for working with such groups or their group elements.

1.2. **Affine varieties.** Reference: [1, §1.1-1.5]

In order to define linear algebraic groups, we first need to explain how to capture the idea of an *algebraic* object. This is done by the notion of an *affine algebraic variety*.

Let k be an algebraically closed field.

Slogan. An affine variety over k is the vanishing set in some k^n defined by a family of polynomial equations.

We now make this definition precise. Throughout, let

$$A \cong k[T_1, \ldots, T_n]/(f_1, \ldots, f_m)$$

be a reduced finite type k-algebra. We want to define the affine variety $\mathrm{Sp}(A)$ attached to A, which will be done in three steps.

1st Step: The underlying set. Define the *Max-Spectrum* of A as

$$\mathrm{Sp}(A) := \{\text{maximal ideals of } A\}.$$

Because k is assumed algebraically closed, for every $\mathfrak{m} \in \mathrm{Sp}(A)$, the residue field of $\mathfrak{m}$ is naturally identified with k:

$$\begin{array}{ccc} k & \longrightarrow & A \\ & \searrow^{\cong} & \downarrow \text{quotient} \\ & & A/\mathfrak{m} \end{array} \qquad \text{(also use } A \text{ is a } k\text{-algebra.)}$$

So we see:

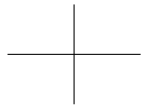
$$\begin{aligned} \mathrm{Sp}(A) &= \{k\text{-alg homom } A \longrightarrow k\} \\ &= \{(x_1, \ldots, x_n) \in k^n \mid f_1(x_1, \ldots, x_n) = \cdots = f_m(x_1, \ldots, x_n) = 0\}. \end{aligned}$$

choosing presentation $A = k[T_1, \ldots, T_n]/(f_1, \ldots, f_m)$

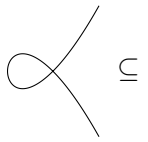
Example 1.10.

$$\mathrm{Sp}(k[T_1,\dots,T_n]) \;=\; k^n$$

$$\mathrm{Sp}(k[X,Y]/(XY)) \;=\; \subseteq k^2$$



$$\mathrm{Sp}(k[X,Y]/(Y^2-X^2(X+1))) \;=\; \subseteq k^2$$



2nd Step: Upgrade to a toplogical space. Every $f \in A$ defines a function $\mathrm{Sp}(A) \longrightarrow k$

$$\begin{array}{ccc} k & \longrightarrow & A \\ & \searrow & \downarrow \mathfrak{m} \in \mathrm{Sp}(A) \\ & & k \end{array} \qquad \begin{array}{c} f \\ \downarrow \\ f \bmod \mathfrak{m}. \end{array}$$

We usually write $x = \mathfrak{m} \in \mathrm{Sp}(A)$ and $f(x) := f \bmod \mathfrak{m} \in k$ where $k = A/\mathfrak{m}$, because this is much more intuitive notation.

Polynomial functions should definitely be continuous, but k has no topology. So the best one can do in algebraic geometry is to *define* open and closed subsets by

$$\begin{aligned} V(f) &= \{x \in \mathrm{Sp}(A) \mid f(x) = 0\} \quad \text{closed "vanishing set"} \\ D(f) &= \{x \in \mathrm{Sp}(A) \mid f(x) \neq 0\} \quad \text{open "doesn't vanish set".} \end{aligned} \tag{1.2}$$

Definition 1.11. Endow $\mathrm{Sp}(A)$ with topology generated by $\{D(f)\}_{f\in A}$.

What does this mean concretely? We already have the following opens from (1.2):

$$\mathrm{Sp}(A) = D(1), \qquad \varnothing = D(0)$$
$$D(f_1) \cap \cdots \cap D(f_r) = D(f_1 \cdots f_r)$$

So the potential new open subsets are the arbitrary unions $\bigcup_{i\in I} D(f_i)$ of sets of the form $D(f)$. Note that an open of the form $D(f)$ is called *principal open subset*.

But A is noetherian: Ther exist finitely many $g_1,\dots,g_r$ generating the ideal $(f_i)_{i\in I}$. Then

$$\bigcap_{i\in I} V(f_i) = \bigcap_{j=1}^{r} V(g_j) \quad \text{or equivalently,} \quad \bigcup_{i\in I} D(f_i) = \bigcup_{j=1}^{r} D(g_j).$$

Conclusion. The open subsets of $\mathrm{Sp}(A)$ are precisely the finite unions $\bigcup_{i=1}^{r} D(f_i)$.

Dually, the closed subsets are precisely the finite intersections

$$V(f_1,\dots,f_r) := \bigcap_{i=1}^{r} V(f_i).$$

Next, we want to connect this topology closer to the ring A. We in fact have back-and-forth definitions:

$$A \supseteq S \text{ subset} \;\longmapsto\; V(S) := \bigcap_{s\in S} V(s),$$
$$I(Z) := \{f \in A \mid f(z) = 0 \;\forall z \in Z\} \;\longleftarrow\; Z \subseteq \mathrm{Sp}(A) \text{ closed.}$$

You should convince yourself of the following easy properties:

- $V(S)$ is closed

- $I(Z)$ an ideal
- $V(S) = V((S))$, $(S) \subseteq A$ ideal generated by S
- If $f^r \in I(Z)$, then $f \in I(Z)$. That is, $I(Z)$ is a so-called *radical* ideal. For example, $V(f^r) = V(f)$.

Theorem 1.12 (Hilbert's Nullstellensatz)**.** *The above maps define a bijection*

$$\begin{aligned} \{\textit{radical } I \subseteq A\} &\overset{1:1}{\longleftrightarrow} \{\textit{closed } Z \subseteq \mathrm{Sp}(A)\} \\ I &\longmapsto V(I), \\ I(Z) &\longmapsfrom Z. \end{aligned}$$

The significance of this theorem is that it allows to translate any statement about unions, intersections, non-emptyness etc. of closed or open subsets of $\mathrm{Sp}(A)$ into a statement about ideals in A, and vice versa. That is, it provides a dictionary between geometry and algebra.

3rd Step: Equip $\mathrm{Sp}(A)$ with its polynomial functions. Set $X = \mathrm{Sp}(A)$ in the following. For every open $U \subseteq X$, we want to say what the polynomial functions on U are. This is similar to the definition of the smooth functions on an open subset of a manifold, for example.

For traditional reasons, what we intuitively just called "polynomial" functions, is usually called *regular functions*.

Definition 1.13. For $U \subseteq \mathrm{Sp}(A)$ open, we denote by $\mathscr{O}_X(U)$ the ring of all regular functions $f : U \to k$.

(1) If $U = D(h)$ is principal open, then by definition

$$\mathscr{O}_X(D(h)) := A[h^{-1}].$$

Note that this makes sense since $h : D(h) \to k^\times$ is an invertible function.

(2) For general U, we say that $f : U \to k$ is regular if $\forall D(h) \subseteq U$, the restriction $f|_{D(h)} \in \mathscr{O}_X(D(h))$.

Definition 1.14. The pair $(\mathrm{Sp}(A), \mathscr{O}_{\mathrm{Sp}(A)})$ is the *affine variety* defined by A. Here, $\mathrm{Sp}(A)$ denotes the topological space, and $\mathscr{O}_{\mathrm{Sp}(A)}$ denotes the regular functions on $\mathrm{Sp}(A)$. But usually one only writes $\mathrm{Sp}(A)$ for brevity.

Example 1.15. The *n-dimensional affine space* over k is

$$\mathbb{A}^n := \mathrm{Sp}\, k[T_1, \ldots, T_n].$$

(1) Underlying set $= k^n$,
(2) Opens = complements of vanishing sets $V(f_1, \ldots, f_m)$,
(3) Regular functions =

$$\mathscr{O}_{\mathbb{A}^n}(U) = \left\{ \frac{p}{q} \in k(T_1, \ldots, T_n) \;\middle|\; q(x) \neq 0 \ \forall x \in U \right\}.$$

Example 1.16. Here is an interesting example of regular functions on a non-principal open: Consider $\mathbb{A}^2 = \mathrm{Sp}\, k[X, Y]$ with open subset $U = k^2 \setminus \{(0,0)\}$. This is the union of $D(X)$ and $D(Y)$. Then

$$\mathscr{O}_{\mathbb{A}^2}(U) = k[X^{\pm 1}, Y] \cap k[X, Y^{\pm 1}] = k[X, Y]$$

where the intersection happens in the **quotient field** $k(X, Y) = \mathrm{Quot}(k[X, Y])$ (also called the **fraction field**).

The above procedure is in fact general: If A is an integral domain, then

$$\mathscr{O}_{\mathrm{Sp}(A)}(U) = \bigcap_{D(f)\subseteq U} A[f^{-1}], \quad \text{intersect in } \mathrm{Quot}(A).$$

It suffices to take the intersection over an open covering $U = D(f_1) \cup \ldots \cup D(f_m)$ in fact.

4th Step: Morphisms. We had three steps to define the object $\mathrm{Sp}(A)$, but we also need to define morphisms between affine varieties. Let $X = \mathrm{Sp}(A)$ and $Y = \mathrm{Sp}(B)$ in the following.

Definition 1.17. A morphism from Y to X is a map $\varphi : Y \to X$ that preserves regular functions. That is, if $V \subseteq Y$ and $U \subseteq X$ are open with $\varphi(V) \subseteq U$, then for every $f \in \mathscr{O}_X(U)$, the composition

$$f \circ \varphi : V \longrightarrow k$$

lies in $\mathscr{O}_Y(V)$.

• In particular, if $\varphi : Y \to X$ is a morphism, we may apply this definition to $V = Y$ and $X = U$ and obtain a ring map

$$\begin{aligned} \varphi^* : A = \mathscr{O}_X(X) &\longrightarrow \mathscr{O}_Y(Y) = B \\ f &\longmapsto f \circ \varphi. \end{aligned}$$

This is the *pullback map* on the coordinate rings of X and Y.

• Conversely, any k-algebra homomorphism $\varphi^* : A \longrightarrow B$ k induces a morphism of affine varieties

$$\begin{aligned} \varphi = \mathrm{Sp}(\varphi^*) : Y &\longrightarrow X \\ B \supseteq \mathfrak{m}_B &\longmapsto (\varphi^*)^{-1}(\mathfrak{m}_B) \subseteq A. \end{aligned}$$

Theorem 1.18 (Hilbert's Nullstellensatz)**.** *The above functors define an equivalence of categories*

$$\begin{aligned} \left\{ \begin{array}{c} \textit{reduced finite type} \\ \textit{k-algebras} \end{array} \right\}^{\mathrm{op}} &\overset{\cong}{\longrightarrow} \left\{ \begin{array}{c} \textit{affine varieties} \\ \textit{over k} \end{array} \right\} \\ A &\longmapsto \mathrm{Sp}(A), \\ \mathscr{O}_X(X) &\longleftarrow\!\shortmid (X, \mathscr{O}_X). \end{aligned}$$

The general principle for passing between maps of varieties and algebra homomorphisms is the following. A morphism

$$k^n \xrightarrow{\varphi} k^m$$

given by an m-tuple of polynomials in n variables

$$(x_1, ..., x_n) \longmapsto (\varphi_1(x_1, ..., x_n), ..., \varphi_m(x_1, ..., x_n))$$

corresponds to the ring map

$$\begin{aligned} k[X_1, ..., X_n] &\xrightarrow{\widetilde{\varphi}} k[T_1, ..., T_m] \\ \varphi_i(X_1, ..., X_m) &\longleftarrow\!\shortmid T_i. \end{aligned}$$

1.3. **Linear algebraic groups.** Reference: [1, §7.1, §7.2]

Throughout, k is an algebraically closed field of characteristic 0. We often abbreviate "linear algebraic group" by LAG.

Definition 1.19. An LAG over k is an affine variety G over k together with a group structure

$$m : G \times G \longrightarrow G$$

which is also a morphism of varieties.

Explanation. Let $X = \mathrm{Sp}(A)$, $Y = \mathrm{Sp}(B)$ be affine varieties over k. Then $X \times Y$ is the affine k-variety $\mathrm{Sp}(A \otimes_k B)$.

Why does this make sense?

- Mathematically:

$$\begin{aligned} A = k[T_1, \ldots, T_n]/(f_1, \ldots, f_m) &\longleftrightarrow X = V(f_1, \ldots, f_m) \subseteq k^n, \\ B = k[S_1, \ldots, S_r]/(g_1, \ldots, g_s) &\longleftrightarrow Y = V(g_1, \ldots, g_s) \subseteq k^r. \end{aligned}$$

$$\implies A \otimes_k B = k[T_1, \ldots, T_n, S_1, \ldots, S_r]/(f_1, \ldots, f_m, g_1, \ldots, g_s)$$

and

$$\mathrm{Sp}(A \otimes_k B) = V(f_1, \ldots, f_m) \times V(g_1, \ldots, g_s) \subseteq k^{n+r},$$

just as desired for a product!

- $\mathrm{Sp}(A \otimes_k B)$ also has the universal property: For $Z = \mathrm{Sp}(C)$, we have

$$\begin{aligned} \mathrm{Mor}(Z, \mathrm{Sp}(A \otimes_k B)) &\underset{\text{Hilbert}}{=} \mathrm{Hom}_{k\text{-alg}}(A \otimes_k B, C) \\ &\underset{-\otimes_k -}{=} \mathrm{Hom}_{k\text{-alg}}(A, C) \times \mathrm{Hom}_{k\text{-alg}}(B, C) \\ &\underset{\text{Hilbert}}{=} \mathrm{Mor}(Z, X) \times \mathrm{Mor}(Z, Y). \end{aligned}$$

$\implies \mathrm{Sp}(A \otimes_k B)$ has univ. prop of $X \times Y$!

Conclusion. If $G = \mathrm{Sp}(A)$, then $G \times G = \mathrm{Sp}(A \otimes_k A)$. Condition on m is that it is defined by polynomials, equivalently that it comes from a map $A \longrightarrow A \otimes_k A$, and that it gives a group structure to the set underlying G.

Example 1.20. The *additive group* $\mathbb{G}_a$.

$\mathbb{G}_a = (k, +)$ with variety structure $k = \mathrm{Sp}(k[t])$.

$$+ : k \times k \longrightarrow k, \qquad (x, y) \longmapsto x + y$$

is the polynomial map corresponding on rings to

$$\begin{aligned} k[x] \otimes k[y] &\xleftarrow{\varphi} k[t] \\ x \otimes 1 + 1 \otimes y &\longleftarrow\!\shortmid\ t. \end{aligned}$$

Example 1.21. The *multiplicative group* $\mathbb{G}_m$.

$\mathbb{G}_m = (k^\times, *)$ with variety structure $k^\times = \mathrm{Sp}(k[t, t^{-1}])$.

$$* : k^\times \times k^\times \longrightarrow k^\times, \qquad (x, y) \longmapsto x \cdot y$$

induced from

$$\begin{aligned} k[x^{\pm 1}] \otimes k[y^{\pm 1}] &\longleftarrow k[t^{\pm 1}] \\ x \otimes y &\longleftarrow\!\shortmid\ t. \end{aligned}$$

Remark 1.22. The above considered $k^\times$ as an open subset of $\mathbb{A}^1$. One can also treat a localization as a closed subset in an affine space of one dimension higher. For $\mathbb{G}_m$, this means identifying

$$k[t_1, t_2]/(t_1 t_2 - 1) \xrightarrow{\sim} k[t, t^{-1}]$$
$$t_1, t_2 \longmapsto t, t^{-1}.$$

In these coordinates, multiplication is obviously given by polynomials:

$$\begin{array}{ccc} k^\times \times k^\times & \xrightarrow{m := *} & k^\times \\ \downarrow & & \downarrow \\ k^2 \times k^2 & \xrightarrow{\widetilde{m}} & k^2 \end{array} \qquad \begin{array}{ccc} (x, y) & \longmapsto & xy \\ \downarrow & & \downarrow \\ (x, x^{-1}, y, y^{-1}) & \longmapsto & (xy, (xy)^{-1}) \end{array}$$

$$\widetilde{m} : (x_1, x_2, y_1, y_2) \longmapsto (x_1 x_2, y_1 y_2).$$

Example 1.23. The *special linear* group SL_n.

$\mathrm{SL}_n = (\mathrm{SL}_n(k), *)$ with variety structure

$$\mathrm{SL}_n = \mathrm{Sp}\big(k[T_{11}, \ldots, T_{nn}]/(\det((T_{ij})_{i,j}) - 1)\big)$$

Here, $\det((T_{ij})_{i,j})$ is the determinant of the matrix with $T_{i,j}$ in the (i,j)-th entry. This is a polynomial in $k[T_{11}, \ldots, T_{nn}]$.

$$\begin{array}{ccc} \mathrm{SL}_n \times \mathrm{SL}_n & \xrightarrow{m := *} & \mathrm{SL}_n \\ \downarrow & & \downarrow \det = 1 \\ k^{n^2} \times k^{n^2} & \xrightarrow{\widetilde{m}} & k^{n^2} \end{array}$$

$$\widetilde{m} : ((x_{ij}), (y_{ij})) \longmapsto \left(\sum_{k=1}^{n} x_{ik} y_{kj} \right)_{i,j}$$

The morphism $\widetilde{m}$ preserves the determinant 1 condition and hence restricts to a multiplication morphism on SL_n.

Example 1.24. The *general linear group* GL_n

$\mathrm{GL}_n = (\mathrm{GL}_n(k), *)$ with variety structure

$$\mathrm{GL}_n = \mathrm{Sp}\big(k[T_{ij}, D]/(D \cdot \det(T_{ij}) - 1)\big) \subseteq k^{n^2} \times k.$$

The relation $D \cdot \det((T_{i,j})_{i,j}) - 1$ forces that D is the inverse of $\det((T_{ij})_{i,j})$, making the determinant invertible. This is the same as localizing at $\det((T_{i,j})_{i,j})$ as in Remark 1.22 before. We have

$$\begin{array}{ccc} \mathrm{GL}_n \times \mathrm{GL}_n & \longrightarrow & \mathrm{GL}_n \\ \downarrow & & \downarrow \\ (k^{n^2} \times k) \times (k^{n^2} \times k) & \longrightarrow & k^{n^2} \times k \end{array} \qquad \begin{array}{c} (x_{ij}) \\ \downarrow \\ ((x_{ij}), \det((x_{ij})_{i,j}) \end{array}$$

$$((x_{ij}), D_x, (y_{ij}), D_y) \longmapsto ((x_{ij}) \cdot (y_{jk}), D_x \cdot D_y).$$

Example 1.25. Let $H \in \mathrm{M}_n(k)$ be a non-degenerate symmetric matrix, i.e. $H^t = H$ and $\det(H) \neq 0$. The *special orthogonal group* defined by H is

$$\mathrm{SO}(H) = \{g \in \mathrm{SL}_n(k) \mid g^t H g = H\} \subseteq \mathrm{SL}_n.$$

Writing out the condition $g^t H g = H$ gives n^2 many quadratic polynomial equations $f_1, \ldots, f_{n^2}$ in the g_{ij}. Correspondingly, the variety structure on $\mathrm{SO}(H)$ is by realizing it as

$$\mathrm{SO}(H) = \mathrm{Sp}\big(k[g_{11}, \ldots, g_{nn}]/(f_1, \ldots, f_{n^2}, \det(g_{ij}) - 1)\big).$$

The group structure is matrix multiplication from SL_n.

Observation. We always assume that k is algebraically closed, so there exists a unique non-degenerate, n variable quadratic form up to isometry.

$\implies$ After change of basis:

$$H = \begin{pmatrix} & \mathbb{1}_m \\ \mathbb{1}_m & \end{pmatrix} \quad n = 2m \quad \text{or} \quad H = \begin{pmatrix} 1 & & \\ & & \mathbb{1}_m \\ & \mathbb{1}_m & \end{pmatrix} \quad n = 2m+1.$$

Here $\mathbb{1}_m$ denotes the $m \times m$ identity matrix. For comparison, recall that over $\mathbb{R}$ a quadratic form is classified by its signature. Correspondingly, there would be several n variable orthogonal groups.

Example 1.26. The *symplectic group* Sp_{2m} over k is defined by

$$\mathrm{Sp}_{2m}(k) = \left\{ g \in \mathrm{M}_{2m}(k) \;\middle|\; g^t \begin{pmatrix} & \mathbb{1}_m \\ -\mathbb{1}_m & \end{pmatrix} g = \begin{pmatrix} & \mathbb{1}_m \\ -\mathbb{1}_m & \end{pmatrix} \right\} \subseteq k^{4m^2}.$$

It is described by $4m^2$ polynomial equations with matrix multiplication just as the special orthogonal group before.

Remark 1.27. $\mathrm{SL}_n, \mathrm{SO}_{2m}, \mathrm{SO}_{2m+1}$ and Sp_{2m} are called the *classical groups*.

2. Topological properties (9/23/2026)

k always denotes an algebraically closed field of characteristic 0.

2.1. **Examples.** Our first goal today is to see more examples of linear algebraic groups and some related easy definitions. Recall that a linear algebraic group G over k is an affine k-variety G together with a multiplication morphism $m : G \times G \to G$ giving its set of points a group structure.

In other words, G can be written as the vanishing locus of a family of polynomials equations $G \subseteq k^n$ in some k^n such that m is induced from a (non-unique) polynomial map $\widetilde{m}$:

$$\begin{array}{ccc} G \times G & \xrightarrow{\quad m \quad} & G \\ \big\downarrow & & \big\downarrow \\ k^n \times k^n & \xrightarrow{\quad \widetilde{m} \quad} & k^n \end{array}$$

Note. Our theory of varieties is built around *reduced* finite type k-algebras. In scheme theory language, we are considering reduced finite-type k-schemes.

Example 2.1. Let H be a symmetric $n \times n$ matrix with $\det(H) \in k^\times$. The *orthogonal group* is

$$O(H) := \left\{ x \in \mathrm{M}_n(k) : x^t H x = H \right\}$$

with variety structure as a closed subvariety of $\mathrm{M}_n(k) \cong \mathbb{A}^{n^2}$. Then, by our conventions, the orthogonal group has the variety structure $O(H) = \mathrm{Sp}(A_{red})$ with

$$A = k[x_{11}, x_{12}, ..., x_{nn}]/(x^t H x - H)$$

where the ideal $(x^t H x - H) \subseteq k[x_{11}, ..., x_{nn}]$ has n^2 many partially redundant equations. It turns out, however, that A is already reduced.

The same method and remark apply to the symplectic group

$$\mathrm{Sp}_{2m} := \left\{ x \in \mathrm{M}_{2m}(k) : x^t \begin{pmatrix} & \mathbb{1}_m \\ -\mathbb{1}_m & \end{pmatrix} x = \begin{pmatrix} & \mathbb{1}_m \\ -\mathbb{1}_m & \end{pmatrix} \right\}$$

The deeper reason for this phenomenon is that *every* finite type group scheme over a field of characteristic 0 is reduced (even smooth). This is the reason for why we work over a field of characteristic 0.

Example 2.2. Let $H = \left(\begin{smallmatrix} & 1 \\ 1 & \end{smallmatrix}\right)$. Then

$$\mathrm{SO}(H) = \left\{x \in \mathrm{M}_2(k) : \det(x) = 1, x^t H x = H\right\}.$$

is regarded as $\mathrm{Sp}(A)$ where $A = k[x_{11}, x_{12}, x_{21}, x_{22}]/I$, here

$$I := (\underbrace{x_{11}x_{22} - x_{12}x_{21} - 1}_{\det(x)=1}, \quad x_{11}x_{22} + x_{12}x_{21} - 1, \quad 2x_{11}x_{21}, \quad 2x_{22}x_{12}).$$

The later three generators of this ideal come from making explicit the $x^t H \overset{!}{=} H$ entry by entry:

$$\begin{aligned}\begin{pmatrix} x_{11} & x_{21} \\ x_{12} & x_{22} \end{pmatrix}\begin{pmatrix} & 1 \\ 1 & \end{pmatrix}\begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix} &= \begin{pmatrix} x_{21} & x_{11} \\ x_{22} & x_{12} \end{pmatrix}\begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix} \\ &= \begin{pmatrix} 2x_{11}x_{21} & x_{12}x_{21} + x_{11}x_{22} \\ x_{12}x_{21} + x_{11}x_{22} & 2x_{22}x_{12} \end{pmatrix} \\ &\overset{!}{=} \begin{pmatrix} & 1 \\ 1 & \end{pmatrix}.\end{aligned}$$

Then, we have the following relations in A:

$$x_{12}x_{21} = -x_{12}x_{21} \implies x_{12}x_{21} = 0 \implies x_{11}x_{22} = 1.$$

In particular, the elements x_{11}, x_{22} are invertible, so $x_{12} = x_{21} = 0$ in A. Hence,

$$A \cong k[x_{11}, x_{22}]/(x_{11}x_{22} - 1),$$

so we get the identification

$$\mathrm{Sp}(A) = \left\{\begin{pmatrix} \lambda & \\ & \lambda^{-1} \end{pmatrix} : \lambda \in k^\times\right\} \subseteq \mathrm{M}_2(k)$$

which is isomorphic to $\mathbb{G}_m$.

Definition 2.3 (Homomorphisms of LAGs)**.** Let G, H be LAGs/k. A *linear algebraic group homomorphism* $f : G \to H$ is a map that is both a group homorphism and morphism of algebraic varieties.

Example 2.4. The determinant defines a homomorphism $\det : \mathrm{GL}_n \to \mathbb{G}_m$,

$$\begin{array}{ccc} \mathrm{GL}_n & \xrightarrow{f := \det} & \mathbb{G}_m \\ \downarrow & & \downarrow \\ k^{n^2} \times k & \xrightarrow{\widetilde{f}} & k \times k \end{array} \qquad \begin{array}{ccc} (x_{ij}) & \longmapsto & \det(x_{ij}) \\ \downarrow & & \downarrow \\ (x_{ij}, \det(x_{ij})^{-1}) & \longmapsto & (\det(x_{ij}), \det(x_{ij})^{-1}). \end{array}$$

The vertical embeddings here are the same as in Examples 1.21 and 1.24. The map corresponding to det on coordinate rings is

$$\begin{aligned} k[x_{ij}, D]/(\det(x_{ij}) \cdot D - 1) &\longleftarrow k[t, t^{-1}] \\ \det(x_{ij}) &\longleftarrow\!\shortmid \quad t \end{aligned}$$

Example 2.5. Let H be a comutative LAG over k, with multiplication $m_H : H \times H \to H$. Then we can multiply $f_1, f_2 : G \to H$ by

$$(f_1 \cdot f_2)(g) = f_1(g) \cdot f_2(g).$$

In this way, $\mathrm{Hom}_{LAG}(G, H)$ becomes a group. Moreover, $\mathrm{End}_{LAG}(H, H)$ is a ring with addition coming from m_H and multiplication being composition.

Exercise. Show that there is an isomorphism $k \xrightarrow{\sim} \mathrm{End}_{LAG}(\mathbb{G}_a)$ given by

$$\alpha \longmapsto [k \mapsto k, x \mapsto \alpha x].$$

In the same way, show that

$$\mathrm{M}_m(k) \xrightarrow{\sim} \mathrm{End}_{LAG}(\mathbb{G}_a^m).$$

Proposition 2.6. *There is an isomorphism of rings*

$$\mathbb{Z} \xrightarrow{\sim} \mathrm{End}_{LAG}(\mathbb{G}_m), \qquad m \longmapsto [x \mapsto x^m].$$

Proof. The variety morphisms $f : \mathbb{G}_m \to \mathbb{G}_m$ correspond to the k-algebra homomorphism $f^* : k[t^{\pm 1}] \to k[t^{\pm 1}]$. The units of $k[t^{\pm 1}]$ are in form of $\alpha \cdot t^m$, for $\alpha \in k^\times$ and $m \in \mathbb{Z}$. Thus

$$f^*(t) = \alpha t^m$$

for unique $\alpha \in k^\times$ and $m \in \mathbb{Z}$, meaning that $f(x) = \alpha x^m$. The group homomorphism condition is equivalent to

$$\begin{aligned} f(xy) = f(x)f(y) &\iff \alpha x^m y^m = \alpha^2 x^m y^m \\ &\iff \alpha = \alpha^2 \\ &\iff \alpha = 1. \end{aligned}$$

Hence $\mathrm{End}_{LAG}(\mathbb{G}_m) = \{x \mapsto x^m \mid m \in \mathbb{Z}\}$. □

Example 2.7 (Inner automorphisms)**.** Let G be a LAG over k and fix $x \in G$. Then

$$G \xrightarrow{\sim} G, \qquad g \longmapsto xgx^{-1},$$

is an automorphism of linear algebraic groups. It is a morphism because it is the composition of

$$G \cong \{x\} \times G \times \{x^{-1}\} \hookrightarrow G \times G \times G \xrightarrow{m \circ (\mathrm{id} \times m)} G.$$

Definition 2.8 (Subgroups of a LAG)**.** Let $X = \mathrm{Sp}(A)$ be an affine variety. A *closed subvariety* of X is a closed subset $Z \subseteq X$, endowed with the variety structure $\mathrm{Sp}(A/I(Z))$.

For a linear algebraic group G, a *subgroup of* G always means a closed subvariety $H \subseteq G$ that is also a subgroup.

Example 2.9. The subgroup $\mathrm{SL}_n \subseteq \mathrm{GL}_n$ is the closed subvariety defined by $\det = 1$:

$$V(\det(x_{ij}) - 1) \subseteq \mathrm{Sp}\big(k[x_{ij}, D]/(\det(x_{ij})D - 1)\big).$$

Example 2.10. The subgroup

$$\left\{ \begin{pmatrix} x_{11} & x_{12} \\ 0 & 1 \end{pmatrix} \;\middle|\; x_{11} \in k^\times,\ x_{12} \in k \right\} \subseteq \mathrm{GL}_2$$

is the closed subvariety defined by $x_{21} = 0$ and $x_{22} = 1$.

Example 2.11. If $H_1, H_2 \subseteq G$ are closed subgroups, then $H_1 \cap H_2$ is a closed subgroup. For example,

$$\left\{ \begin{pmatrix} x_{11} & x_{12} \\ 0 & x_{22} \end{pmatrix} \right\} \cap \mathrm{SL}_2 = \left\{ \begin{pmatrix} x_{11} & x_{12} \\ 0 & x_{11}^{-1} \end{pmatrix} \right\},$$

where the first set is viewed inside GL_2. The intersection is closed and can be defined by $x_{21} = 0$ and $x_{11}x_{22} = 1$.

Example 2.12 (Kernels are closed subgroups)**.** Let $f : G \to H$ be a homomorphism of linear algebraic groups. Since f is continuous and $\{e_H\} \subseteq H$ is closed,

$$\ker(f) = f^{-1}(e_H)$$

is a closed subgroup of G.

We generalize the definition of kernels using fiber products. First recall the definition:

Definition 2.13. Given maps of sets $X \xrightarrow{f} S \xleftarrow{g} Y$, their *fibre product* is

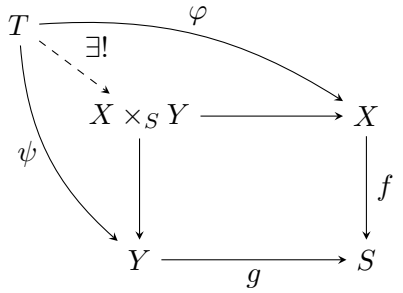
$$X \times_S Y = \{(x, y) \in X \times Y \mid f(x) = g(y)\}.$$

In other words, for every $s \in S$, we take the product of the fibres $f^{-1}(s) \times g^{-1}(s) \subseteq X \times Y$.

Universal property. The fiber product has the following property: For every set T,

$$\mathrm{Map}(T, X \times_S Y) = \{(\varphi, \psi) \in \mathrm{Map}(T, X) \times \mathrm{Map}(T, Y) \mid f \circ \varphi = g \circ \psi\}.$$

This is better memorized by the following diagram, where the condition is that the outer square is commutative:



Instead of working with just sets, this whole construction also works in the category of affine varieties over k. Consider

$$S = \mathrm{Sp}(A), \qquad X = \mathrm{Sp}(B), \qquad Y = \mathrm{Sp}(C),$$

with morphisms $f\colon X \to S$ and $g\colon Y \to S$.

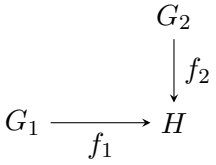
Proposition 2.14. *The affine variety* $\mathrm{Sp}\big((B \otimes_A C)_{\mathrm{red}}\big)$ *is a fibre product of* X *and* Y *over* S.

Proof. Let D be any reduced k-algebra of finite type. Then

$$\begin{aligned}
&\mathrm{Mor}_{k\text{-var}}\big(\mathrm{Sp}(D), \mathrm{Sp}((B \otimes_A C)_{\mathrm{red}})\big) \\
&\quad = \quad \mathrm{Mor}_{k\text{-alg}}\big((B \otimes_A C)_{\mathrm{red}}, D\big) \\
&\quad \overset{D \text{ reduced}}{=} \mathrm{Mor}_{k\text{-alg}}(B \otimes_A C, D) \\
&\quad = \quad \big\{\ (\varphi^*\colon B \to D,\ \psi^*\colon C \to D) \ \big|\ \varphi^* \circ f^* = \psi^* \circ g^*\ \big\} \\
&\quad = \quad \big\{\ (\varphi\colon \mathrm{Sp}(D) \to X,\ \psi\colon \mathrm{Sp}(D) \to Y) \ \big|\ f \circ \varphi = g \circ \psi\ \big\}.
\end{aligned}$$

□

Now assume all the morphisms involved are homomorphisms of linear algebraic groups:



Then

$$G = G_1 \times_H G_2 \subseteq G_1 \times G_2$$

is a LAG and a closed subgroup of $G_1 \times G_2$. For $G_1 = \{e_H\} \hookrightarrow H$, this recovers $\ker(f_2)$.

Example 2.15. The fibre-product square

$$\begin{array}{ccc} G & \longrightarrow & \mathrm{GL}_n \\ \downarrow & \square & \downarrow \det \\ \mathrm{GL}_n & \xrightarrow[\det]{} & \mathbb{G}_m \end{array}$$

gives

$$G = \big\{(g_1, g_2) \in \mathrm{GL}_n \times \mathrm{GL}_n \mid \det(g_1) = \det(g_2)\big\}.$$

2.2. **Topological properties.** Reference: [1, §§1.1–1.3].

In this section, we will study the three topological properties of being quasi-compact, irreducible, and connected.

2.2.1. *Quasi-compactness.*

Definition 2.16. A topological space X is called *quasi-compact* if every open covering $X = \bigcup_{i\in I} U_i$ has a finite subcover.

Proposition 2.17. *For every reduced k-algebra A of finite type, $X = \mathrm{Sp}(A)$ is quasi-compact.*

Proof. Let $X = \bigcup_{i\in I} U_i$ be an open covering of X. We first cover each U_i by principal open subsets $D(f_{ij})$, $j \in J_i$, using that these form a basis of the topology. Clearly, if we can even find finitely many $D(f_{i_1,j_1}), \ldots, D(f_{i_r,j_r})$ covering X, then larger $U_{i_1}, \ldots, U_{i_r}$ also cover X. So we may assume all $U_i = D(f_i)$ principal to begin with.

Recall that $D(f) \subseteq \mathrm{Sp}(A)$ is the subset of maximal ideals $\mathfrak{m} \subseteq A$ not containing f. So $X = \bigcup_{i\in I} D(f_i)$ means that no maximal ideal of A contains all f_i. In other words, the ideal $(f_i)_{i\in I}$ generated by the f_i is not contained in any maximal ideal. This means it equals the unit ideal. So there exist indices $i_1, \ldots, i_r$ and a way to write

$$1 = h_1 f_{i_1} + \ldots + h_r f_{i_r}.$$

Then $X = D(f_{i_1}) \cup \ldots \cup D(f_{i_r})$ is the desired finite subcovering. □

Remark 2.18. This proof did not use any Noetherian property, but the following exercise does.

Exercise. Let $U \subseteq X$ be any open subset. Using that A is noetherian, show that $Z = X \setminus U$ is the vanishing set of finitely many $f_1, \ldots, f_r \in A$. Deduce that the $D(f_i)$ form an open covering of U. Deduce further that U is quasi-compact.

2.2.2. *Irreducibility.*

Definition 2.19. A topological space X is *irreducible* for all closed $Z_1, Z_2 \subseteq X$ with $X = Z_1 \cup Z_2$, $Z_1 = X$ or $Z_2 = X$.

A subset $Z \subseteq X$ is irreducible if it is irreducible with respect to the subspace topology. This means that whenever $Z \subseteq Z_1 \cup Z_2$ for two closed subsets $Z_1, Z_2 \subseteq X$, then already $Z \subseteq Z_1$ or $Z \subseteq Z_2$.

Proposition 2.20. *Let $X = \mathrm{Sp}(A)$. There is a bijection*

$$\begin{aligned} \{\textit{prime ideals } \mathfrak{p} \subset A\} &\overset{1:1}{\longleftrightarrow} \{\textit{nonempty irreducible closed } Z \subseteq X\}, \\ \mathfrak{p} &\longmapsto V(\mathfrak{p}), \\ I(Z) &\longmapsto\!\shortmid\; Z. \end{aligned}$$

Proof. Let $\varnothing \neq Z \subseteq X$ be closed and irreducible, and suppose $fg \in I(Z)$. Then

$$Z \subseteq V(f) \cup V(g).$$

By irreducibility, $Z \subseteq V(f)$ or $Z \subseteq V(g)$. Thus $f \in I(Z)$ or $g \in I(Z)$, so $I(Z)$ is a prime ideal.

Conversely, let $\mathfrak{p}$ be a prime ideal, and suppose that $V(\mathfrak{p}) \subseteq Z_1 \cup Z_2$ for closed subsets Z_1, Z_2. Assume that $V(\mathfrak{p}) \nsubseteq Z_1$ and $V(\mathfrak{p}) \nsubseteq Z_2$. Then by Hilbert's Nullstellensatz, there are functions

$$f_i \in I(Z_i) \smallsetminus \mathfrak{p} \qquad (i = 1, 2),$$

The product $f_1 f_2$ vanishes on $Z_1 \cup Z_2$, hence on $V(\mathfrak{p})$. Again by Hilbert's Nullstellensatz, $f_1 f_2 \in \mathfrak{p}$. Since $\mathfrak{p}$ is prime, this implies $f_1 \in \mathfrak{p}$ or $f_2 \in \mathfrak{p}$, a contradiction. □

Proposition 2.21. *Any closed subset $Z \subseteq X = \mathrm{Sp}(A)$ is a finite union of irreducible closed subsets.*

Proof. Suppose there is a closed subset $Z_0 \subseteq X$ that cannot be written as a finite union of irreducible closed subsets. In particular, Z_0 is not irreducible, so

$$Z_0 = Z_1 \cup Z_1', \qquad Z_1 \subsetneq Z_0, \quad Z_1' \subsetneq Z_0,$$

and at least one of Z_1 and Z_1' cannot be written as a finite union of irreducible closed subsets. Say this is Z_1. Continuing in this way, we obtain a strictly descending chain

$$Z_0 \supsetneq Z_1 \supsetneq Z_2 \supsetneq \cdots.$$

By Hilbert's Nullstellensatz, this corresponds to a *strictly* ascending chain of ideals

$$\mathfrak{a}_0 \subsetneq \mathfrak{a}_1 \subsetneq \mathfrak{a}_2 \subsetneq \cdots, \qquad \mathfrak{a}_i = I(Z_i).$$

The strictness of the inclusions is essential here and comes from the bijectivity part of Hilbert's Nullstellensatz. But A is Noetherian, so this chain must stabilize, contradicting our original assumptions. □

Corollary 2.22. *Any closed $Z \subseteq \mathrm{Sp}(A)$ can be written as a finite union*

$$Z = Z_1 \cup \cdots \cup Z_r, \qquad Z_i \nsubseteq Z_j \quad \text{for } i \neq j,$$

of closed irreducible subsets. The Z_i are called the irreducible components *of Z. They are in bijection as*

$$\{\textit{irreducible components of } Z\} \overset{1:1}{\longleftrightarrow} \left\{ \begin{matrix} \textit{minimal prime ideals} \\ \textit{of } A \textit{ containing } I(Z) \end{matrix} \right\}. \tag{2.1}$$

The corollary of course also applies to $\mathrm{Sp}(A)$ itself, and in fact is equivalent to this case by $Z = \mathrm{Sp}(A/I(Z))$.

2.2.3. *Connectedness.*

Definition 2.23. A topological space X is *connected* if every subset $U \subseteq X$ that is both open and closed is either $\varnothing$ or X. Equivalently, we cannot write

$$X = U_1 \sqcup U_2$$

as a disjoint union of two nonempty open subsets.

Lemma 2.24. *Let $X = \mathrm{Sp}(A)$. There is a bijection*

$$\{U \subseteq X \mid U \text{ is open and closed}\} \longleftrightarrow \{e \in A \mid e^2 = e\},$$

between open and closed subsets and idempotents in A, given by

$$\begin{array}{rcl} U & \longmapsto & \mathbb{1}_U \in \mathscr{O}(X), \\ D(e) & \longleftarrow\!\shortmid & e. \end{array}$$

Here $\mathbb{1}_U$ denotes the indicator function of U.

Proof. We first show that $\mathbb{1}_U$ is regular, i.e. that it lies in $\mathscr{O}(X)$. Since U and U^c are both open, we can cover them by principal opens as

$$U = \bigcup_i D(f_i), \qquad U^c = \bigcup_j D(g_j).$$

Then

$$\mathbb{1}_U|_{D(f_i)} = 1 \in A[f_i^{-1}], \qquad \mathbb{1}_U|_{D(g_j)} = 0 \in A[g_j^{-1}].$$

Thus, taking our definition of regularity (Definition 1.13), we get that $\mathbb{1}_U$ is regular as claimed. By the mutual quasi-inverse part of the strong form of Hilbert's Nullstellensatz (Theorem 1.18), this means $\mathbb{1}_U$ equals (the function defined by) some $e \in A$. The map $U \mapsto \mathbb{1}_U$ is clearly injective and $\mathbb{1}_U^2 = \mathbb{1}_U$.

Conversely, if $e^2 = e$, then $e(x)^2 = e(x)$ for every $x \in X$. Thus $e(x) \in \{0, 1\}$ and

$$X = D(e) \sqcup D(1 - e).$$

This shows that $D(e)$ is both open and closed, so the map from right to left is well defined. The two maps are clearly mutually inverses. □

Exercise. (a) Let $X_1, X_2 \subseteq X$ be two subsets of a topological space which are connected with respect to the subspace topology. Show that if $X_1 \cap X_2 \neq \emptyset$, then $X_1 \cup X_2$ is connected as well.

(b) Specialize to $X = \mathrm{Sp}(A)$ and its irreducible components $X_1, \ldots, X_r$. Show that the X_i can be uniquely grouped by $\{1, \ldots, n\} = I_1 \sqcup \ldots \sqcup I_k$ such that each union $\bigcup_{i \in I_j} X_i$ is connected, open and closed in X. These unions are called the *connected components* of X.

The exercise in particular proves the following proposition:

Proposition 2.25. *Let A be a finite type k-algebra. Then A has only finitely many idempotents.*

Algebraically, we can uniquely up to isomorphism decompose

$$A \cong A_1 \times \cdots \times A_k$$

as a product such that $0, 1 \in A_i$ are the only idempotents. Geometrically, this corresponds to a disjoint union decomposition

$$\mathrm{Sp}(A) = \mathrm{Sp}(A_1) \sqcup \cdots \sqcup \mathrm{Sp}(A_k)$$

with each $\mathrm{Sp}(A_i)$ is open, closed, and connected. Each $\mathrm{Sp}(A_i)$ corresponds to a union $\bigcup_{i \in I_j} X_i$ of irreducible components in the terminology of the above exercise.

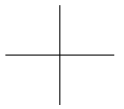
Example 2.26. If A is an integral domain, then $\mathrm{Sp}(A)$ is connected, because the only solutions of $e(e - 1) = 0$ are $e = 0$ and $e = 1$.

Remark 2.27. The real picture can sometimes be deceptive:

$$\mathrm{Sp}(k[t^{\pm 1}]) \quad = \quad$$

is connected as $k[t, t^{-1}]$ is an integral domain. It is better to think of $\mathbb{C} \smallsetminus \{0\}$ which is connected.

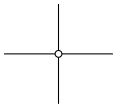
Example 2.28. Let $A = k[x, y]/(xy)$. Geometrically,

$$\mathrm{Sp}(A) \quad = \quad$$  $$\quad \subseteq \quad k^2.$$

There are two irreducible components:

$$\mathrm{Sp}(A) = V(x) \cup V(y).$$

Consider $x + y \in A$. It vanishes only at $(0, 0)$, so

$$\mathrm{Sp}\big(A[(x+y)^{-1}]\big) \quad = \quad$$  $$\quad .$$

In $A[(x + y)^{-1}]$, we calculate

$$\left(\frac{x}{x+y}\right)^2 = \frac{x^2}{x^2 + 2xy + y^2} = \frac{x^2}{x^2 + y^2} = \frac{x}{x+y},$$

because $x(x^2 + y^2) = x^2(x + y)$ in A (clear denominators). Therefore, $x/(x + y)$ and similarly $y/(x + y)$ are idempotents in $A[(x + y)^{-1}]$. This is clear geometrically, because $x/(x + y)$ is 1 on the axis $\{y = 0\}$ and 0 on the axis $\{x = 0\}$.

One checks that this in fact gives an isomorphism

$$A[(x + y)^{-1}] \cong k[x, x^{-1}] \times k[y, y^{-1}],$$

which describes the two connected components of $\mathrm{Sp}(A[(x+y)^{-1}]$ in line with Proposition 2.25.

2.3. **Connectedness for LAGs.** Reference: [1, §7.3].

We now want to study the topological properties of linear algebraic groups. First, we need a lemma.

Lemma 2.29. *Let X and Y be irreducible affine k-varieties. Then $X \times Y$ is again irreducible.*

Proof. Suppose that $X \times Y = Z_1 \cup Z_2$ is a union of two closed subsets. For every $x \in X$, the subset $\{x\} \times Y$ is closed, and by irreducibility of Y we have

$$\{x\} \times Y \subseteq Z_1 \quad \text{or} \quad \{x\} \times Y \subseteq Z_2.$$

Define

$$X_i = \{x \in X \mid \{x\} \times Y \subseteq Z_i\} \qquad (i = 1, 2).$$

Then $X = X_1 \cup X_2$ by the argument just given. We may also write X_i as the (usually infinite) intersection

$$X_i = \bigcap_{y \in Y} (X \times \{y\} \cap Z_i)\,.$$

This shows that X_i is closed. Since X is irreducible, either $X = X_1$ or $X = X_2$. Therefore, either $X \times Y = Z_1$ or $X \times Y = Z_2$. □

Lemma 2.30. *There is a unique irreducible component $G^\circ \subseteq G$ containing the identity e.*

Proof. Let $X_1, \dots, X_r \subseteq G$ be the irreducible components containing e. By Lemma 2.29, $X_1 \times \cdots \times X_r$ is irreducible. Consider the multiplication morphism

$$m^{(r)} : G^r \longrightarrow G.$$

The image of an irreducible topological space under a continuous map is irreducible (exercise). Hence

$$Y := m^{(r)}(X_1 \times \cdots \times X_r) \subseteq G$$

is irreducible. Since $e \in Y$, we have $Y \subseteq X_i$ for some i. But $X_j \subseteq Y$ for every j and $X_i \not\subseteq X_j$ whenever $i \neq j$. So the only possibility is $r = 1$, proving the lemma. □

Proposition 2.31. *The identity component $G^\circ \subseteq G$ is an open normal subgroup. Its cosets are the irreducible components of G, which are simultaneously the connected components of G.*

Proof. • By Lemma 2.29, $G^\circ \times G^\circ$ is irreducible, so $m(G^\circ \times G^\circ)$ is contained in an irreducible component of G. Since $e \in m(G^\circ \times G^\circ)$, this compenent has to be G°. Moreover, $m(g, e) = g$ for every $g \in G^\circ$, and therefore

$$m(G^\circ \times G^\circ) = G^\circ.$$

Thus multiplication restricts to a morphism

$$m : G^\circ \times G^\circ \longrightarrow G^\circ.$$

• For every $x \in G$, left translation

$$L_x : G \longrightarrow G, \qquad g \longmapsto xg,$$

is a morphism because it is the composition of two morphism:

$$\{x\} \times G \longrightarrow G \times G \xrightarrow{m} G.$$

It is even an isomorphism with inverse $g \mapsto x^{-1}g$. For each $x \in G^\circ$, the translate $x^{-1}G^\circ$ is therefore again an irreducible component of G. It contains e because $x \in G^\circ$, so $x^{-1}G^\circ = G^\circ$. In particular, $x^{-1} = x^{-1} \cdot e \in G^\circ$ which means that G° is stable under $x \mapsto x^{-1}$ and hence a closed subgroup.

• Since G is an affine variety, it has finitely many irreducible components. Write

$$G = G^\circ \cup \bigcup_{i=1}^{r} X_i, \qquad \text{each } X_i \neq G^\circ, \quad X_i \neq X_j \text{ for } i \neq j.$$

If $x \in X_i$, translation by x is an isomorphism of G. In particular, $x^{-1}X_i$ is an irreducible component containing e, so $x^{-1}X_i = G^\circ$. Thus $X_i = xG^\circ$, and all irreducible components are cosets of G°.

• Different cosets are disjoint, i.e. $xG^\circ \cap yG^\circ = \varnothing$ whenever $xG^\circ \neq yG^\circ$. So for each xG°, we can write

$$xG^\circ = G \smallsetminus \bigcup_{yG^\circ \neq xG^\circ} yG^\circ.$$

The union on the right is a finite union of closed subsets, hence closed. Thus the complement xG° is open.

• Finally, for normality, consider any $x \in G$. Conjugation

$$c_x : G \longrightarrow G, \qquad g \longmapsto xgx^{-1},$$

is an automorphism with inverse $c_{x^{-1}}$ (Example 2.7). Hence $xG^\circ x^{-1}$ is an irreducible component again. Since it contains e, it equals G°. □

We end this section with some examples.

Example 2.32. The general linear group GL_n is the principal open subset $D(\det(x_{ij}))$ in $\mathrm{M}_n(k)$. In particular, its coordinate ring

$$k[\mathrm{GL}_n] := \mathscr{O}_{\mathrm{GL}_n}(\mathrm{GL}_n)$$

is the localization $k[x_{11}, \ldots, x_{nn}, \det(x_{ij})^{-1}]$ and hence an integral domain. It follows that GL_n is irreducibe and connected, see (2.1). Therefore, $\mathrm{GL}_n^\circ = \mathrm{GL}_n$.

Example 2.33. Forgetting about the group structure on GL_n, i.e. viewing it solely as an affine variety, there exists an isomorphism

$$\begin{aligned} \mathrm{GL}_n &\xleftarrow{\sim} \mathbb{G}_m \times \mathrm{SL}_n \\ \begin{pmatrix} \lambda & \\ & \mathbb{1}_{n-1} \end{pmatrix} \cdot (x_{ij}) &\longmapsto (\lambda, (x_{ij})). \end{aligned}$$

In general, a product of (non)-irreducible affine varieties is (non)-irreducible, hence SL_n is irreducible. That is, $\mathrm{SL}_n^\circ = \mathrm{SL}_n$.

Example 2.34. Given $k \geq 1$, consider the homomorphism

$$\varphi : \mathrm{GL}_n \longrightarrow \mathbb{G}_m, \qquad x \longmapsto \det(x)^k.$$

Its kernel is

$$\ker(\varphi) = \{x \in \mathrm{GL}_n \mid \det(x) \in \mu_k\}$$

where $\mu_k := \{1, \zeta, \ldots, \zeta^{k-1}\}$ is group of k-th roots of unity. By the previous example, its identity component is $\ker^\circ(\varphi) = \mathrm{SL}_n$ and $\ker(\varphi)/\ker(\varphi)^\circ \xrightarrow{\sim} \mu_k$.

Example 2.35. Consider a non-degenerate symmetric form $H \in \mathrm{Sym}_n(k)$ and the determinant homomorphism

$$\det : \mathrm{O}(H) \longrightarrow\!\!\!\!\!\rightarrow \{\pm 1\}.$$

Its kernel is $\mathrm{SO}(H)$, hence

$$\mathrm{O}(H)/\mathrm{SO}(H) \xrightarrow{\sim} \{\pm 1\}.$$

We see that $\mathrm{O}(H)^\circ \subseteq \mathrm{SO}(H)$, so $\mathrm{O}(H)$ is not connected.

Non-obvious fact: $\mathrm{SO}(H)$ is connected. Hence $O(H)^\circ = \mathrm{SO}(H)$.

In the same way, Sp_{2m} is connected.

3. Embedding into GL_n and images of morphisms (9/30/2026)

Let $\varphi : G \to H$ be a homomorphism of LAGs. We saw as an easy example last week that $\ker(\varphi) \subseteq G$ is a closed subgroup and, in particular, naturally again an LAG (Example 2.12). Today, our goal is to prove that also the image $\mathrm{Im}(\varphi) \subseteq H$ is a closed subgroup, which is at a whole different difficulty level. We need two new tools as inputs:

(1) We will show that for any LAG G, the map $G \to G$, $x \mapsto x^{-1}$ is a morphism of varieties. We will obtain this as a corollary to a fundamental theorem of the theory: Every LAG can be embedded as a closed subgroup into some GL_n.

(2) We will prove that images of variety morphisms are *constructible*. This is a basic result from algebraic geometry which we will discuss in the middle of the lecture.

Remark 3.1. Property (1) has to be imposed as axiom when working with potentially non-reduced group objects such as group schemes or not geometrically reduced varieties. It is also required from the beginning in [1, §7.1].

In our course, we started from slightly weaker axioms. We only assumed that G is an affine variety and that $m : G \times G \to G$ is a morphism of varieties that induces a group structure on G viewed as set. So, for example, we currently do not even know that $x \mapsto x^{-1}$ is a continuous map.

3.1. **Embedding into GL_n.** Reference: [1, §8.6].

Theorem 3.2. *Let G/k be a LAG. There exist $n \in \mathbb{N}$ and an embedding as a closed subgroup*

$$G \overset{\varphi}{\hookrightarrow} \mathrm{GL}_n.$$

Here, by *embedding* we more precisely mean a closed immersion of affine varieties (i.e. taking a quotient on coordinate rings).

Corollary 3.3. *The inversion map $i : G \to G$, $x \mapsto x^{-1}$ is a variety automorphism.*

Proof. Choose an embedding $\varphi : G \hookrightarrow \mathrm{GL}_n$ as in Theorem 3.2. The inversion map

$$i : \mathrm{GL}_n \longrightarrow \mathrm{GL}_n, \qquad x \longmapsto x^{-1}$$

is a morphism because it is given by the polynomial formula $x^{-1} = \det(x)^{-1} \cdot x^{\mathrm{ad}}$ where x^{ad} is the adjunct matrix. Inversion on GL_n preserves the closed subset $\varphi(G) \subset \mathrm{GL}_n$ because G is a group and φ a homomorphism, so it induces an automorphism on the closed subvariety $\varphi(G)$. Since $i : G \xrightarrow{\sim} \varphi(G)$ by what it means to be a closed embedding, inversion restricts to an automorphism $i : G \longrightarrow G$ which is just $x \mapsto x^{-1}$. □

Goal. Prove Theorem 3.2 for a given LAG (G, m). Recall our three axioms:

- G is an affine k-variety
- $m : G \times G \longrightarrow G$ is a morphism of k-varieties
- m induces group structure on G viewed as set.

Translation. Recall that $k[G]$ is the ring consisting of the regular functions $f : G \to k$. For $x \in G$, we define the left x-translate of f by

$$(\tau_x f)(y) := f(x^{-1}g).$$

We choose this convention because $\tau_x(\tau_y(f)) = \tau_{xy}(f)$. That is, $x \mapsto \tau_x$ is a left representation of G on $\mathrm{Map}(G, k)$.

Claim 0. For $x \in G$ and $f \in k[G]$, the translate $\tau_x(f)$ again lies in $k[G]$.

Proof. Consider the composition

$$\psi_x : \{x^{-1}\} \times G \hookrightarrow G \times G \overset{m}{\longrightarrow} G.$$

This is a composition of two morphisms, hence a morphism. So the pullback $\tau_x(f) = f \circ \psi_x$ is again a regular function on G. □

Claim 1. Let $x \in G$ and $f \in k[G]$. There exists a finite dimensional k-subvector space $F \subseteq k[G]$ containing f such that F is stable under left translation by all $x \in G$. That is, for all functions $h \in F$ and $x \in G$, $\tau_x(h)$ again lies in F.

Proof. Consider the pullback $m^*(f) = f \circ m$. It is a regular function on $G \times G$, i.e. lies in $k[G \times G] = k[G] \otimes_k k[G]$. Explicitly,

$$f(x, y) = f(xy).$$

Any element of $k[G] \otimes_k k[G]$ is a finite sum of elementary tensors, so there exist $a_1, \ldots, a_r$, $h_1, \ldots, h_r \in k[G]$ such that for all $x, y \in G$,

$$f(xy) = \sum_{i=1}^{r} a_i(x) \otimes h_i(y).$$

Then, for every $x \in G$, we get that

$$\tau_x(f) = \sum_{i=1}^{r} a_i(x^{-1})h_i$$

is a linear combination of $h_1, \dots, h_r$. So the k-vector space $F_f := \operatorname{span}_k\{\tau_x(f)\}_{x\in G}$ spanned by all left translates

- is finite-dimensional because it is contained in $\operatorname{span}_k\{h_1, \dots, h_r\}$,
- contains $f = \tau_e(f)$ and is stable under left translation because $\tau_z(\tau_x(f)) = \tau_{zx}(f)$.

That is, F_f fulfilles all requirements of Claim 1. □

Claim 2. Assume $F \subseteq k[G]$ is a (finite or infinite-dimensional) k-subvector space that is stable under all left translations. Then $f \circ m \in k[G] \otimes_k F$ for all $f \in F$.

Proof. Choose a basis $\{f_i\}_{i\in I}$ for F. Complete it to a basis $\{f_i\}_{i\in I} \sqcup \{h_j\}_{j\in J}$ of $k[G]$. For every $f \in F$, we can then write

$$f \circ m = \sum_{i\in I} a_i \otimes f_i + \sum_{j\in J} b_j \otimes h_j$$

for unique $a_i, b_j \in A$. (Only finitely many a_i and b_j are nonzero.) Specializing this identity along any

$$G \xrightarrow{(x,\mathrm{id})} G \times G$$

gives

$$\tau_x^{-1}(f) = \sum_{i\in I} a_i(x)f_i + \sum_{j\in J} b_j(x)h_j. \tag{3.1}$$

But $\tau_x^{-1}(f)$ lies in F by assumption, and any linear combination as in (3.1) is unique because the f_i and h_j form a basis for $k[G]$, so we get $b_j(x) = 0$ for all $j \in J$ and $x \in G$. In other words, $b_j : G \to k$ is the zero map. This means that $f \circ m \in k[G] \otimes_k F$ as claimed. □

Proof of Theorem 3.2. Starting with G, let $f_1, \dots, f_r \in k[G]$ be k-algebra generators. For each, let F_i be a finite-dimensional translation invariant space containing it as in Claim 1. For example, take $F_i = F_{f_i}$. Then $F = \sum_{i=1}^{r} F_i$ is still finite-dimensional, translation invariant, and contains $f_1, \dots, f_r$. We obtain a homomorphism

$$\begin{aligned} G &\longrightarrow \mathrm{GL}(F) \\ x &\longmapsto \tau_x \end{aligned} \tag{3.2}$$

and will show that this defines a closed embedding of LAGs.

For concreteness, we first choose a k-basis $h_1, \dots, h_n$ for F. By Claim 2, $m^*(F) \subset k[G]\otimes F$, so there exist unique functions $m_{ji} \in k[G]$ such that

$$m^*(h_i) = \sum_{j=1}^{n} m_{ji} \otimes h_j. \tag{$\star$}$$

For every $x \in G$, this defines a matrix $\varphi'(x) = (m_{ij}(x))_{i,j} \in M_n(k)$. Evaluating ($\star$) at $(x, y) \in G \times G$, we see

$$h_i(xy) = \sum_{j=1}^{n} m_{ji}(x)h_j(y).$$

That is, $\varphi'(x)$ expresses the right representation $f \mapsto (\tau_x^{-1})(f)$ on F as matrix acting from the left in the basis $h_1, \dots, h_n$. So we have the right action rule

$$\varphi'(xy) = \varphi'(y)\varphi'(x).$$

But elements of G are invertible, so φ' factors through $\mathrm{GL}_n \subseteq \mathrm{M}_n(k)$. We can compose it with the inverse and define $\varphi(x) := \varphi'(x)^{-1}$. Since taking the inverse on GL_n is a morphism (see proof of Corollary 3.3 above), and since $(ab)^{-1} = b^{-1}a^{-1}$, this defines a homomorphism of LAGs

$$\varphi : G \longrightarrow \mathrm{GL}_n.$$

Clearly, $\varphi = \tau_x$ as in (3.2).

Final claim. The $\{m_{ji}\}_{i,j}$ generate $k[G]$ as k-algebra. In particular, φ is a closed immersion.

Proof. We use the identity axiom

$$\begin{array}{ccc} G & \xrightarrow{(\mathrm{id}, e)} & G \times G \\ & \searrow\!\!\searrow & \downarrow m \\ & & G \end{array}$$

which when applied to $(\star)$ gives

$$h_i = h_i \circ (\mathrm{id}, e) \circ m = \sum_{j=1}^{n} m_{ji} h_j(e).$$

Hence the $h_1, \dots, h_n$ lie in the space generated by the $\{m_{ji}\}$, and since the $h_1, \dots, h_n$ spanned F which contains generators for $k[G]$, this means that the m_{ji} generate $k[G]$ as well. □

This concludes the proof of the theorem. □

Remark 3.4. Theorem 3.2 also explains where the term "linear" in LAG comes from:

$$\text{LAGs} \;=\; \text{the closed subgroups of general linear groups.}$$

3.2. **Varieties.** We take a moment to introduce general varieties. This investment will pay off in the future when we study group actions. Today, it will only come up as a more natural setting for constructible sets.

Definition 3.5. A *variety* over k is a pair

$$(X, \mathscr{O}_X)$$

where X is topological space, and where $\mathscr{O}_X$ is the following datum: For every open $U \subseteq X$, a subring

$$\mathscr{O}_X(U) \subseteq \mathrm{Map}(U, k).$$

Here, $\mathrm{Map}(U, k)$ is a ring with $(f + g)(x) = f(x) + g(x)$ and $(fg)(x) = f(x)g(x)$. The function in $\mathscr{O}_X(U)$ are called the *regular* functions on U. We require that $(X, \mathscr{O}_X)$ satisfies the following three axioms:

• Whether or not a function $f : U \to k$ is regular can be checked locally. That is, for all open coverings $U = \bigcup_{i \in I} U_i$,

$$f \text{ is regular} \quad \Longleftrightarrow \quad \text{each restrection } f|_{U_i} \text{ is regular.}$$

• Locally, $(X, \mathscr{O}_X)$ is isomorphic to an affine variety. That is, every $x \in X$ has an open neighborhood U such that

$$(U, \mathscr{O}_X|_U) \cong (\mathrm{Sp}(A), \mathscr{O}_{\mathrm{Sp}(A)})$$

for some reduced finite type k-algebra A.

• The topological space X is quasi-compact. That is, there exists a finite open covering of X by opens subsets $\mathrm{Sp}(A_1), \ldots, \mathrm{Sp}(A_r)$.

Remark 3.6. Our notion of variety is equivalent to reduced finite type k-scheme. Traditionally, "variety" also often includes the condition that X is irreducible, but we will work with the more flexible notion.

Exercise. Extend the proof of Proposition 2.21 to show that every closed subset of a variety is a finite union of irreducible subsets. *Hint:* First show that every descending chain of closed subsets of a variety stabilizes.

Definition 3.7. A *morphism of varieties* $\varphi : (Y, \mathscr{O}_Y) \to (X, \mathscr{O}_X)$ is a continuous map $\varphi : Y \to X$ which preserves regular functions. That is, for every two open subsets $U \subseteq X$ and $V \subseteq Y$ with $\varphi(V) \subseteq U$, and for every regular function $f : U \to k$, the composition

$$V \xrightarrow{\varphi} U \xrightarrow{f} k$$

is again regular.

Trivial example: An open or closed subsets of a variety again has a natural variety structure. These are the *open* resp. *closed* subvarieties.

Definition 3.8. A subset $W \subseteq X$ of a topological space is called *locally closed* if it can be written as the intersection of an open and a closed subsets. (It is closed in an open subset, hence the terminology.)

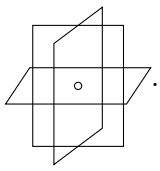
If $(X, \mathscr{O}_X)$ is a variety, then every locally closed subset $W = U \cap Z$ has a natural variety structure $(W, \mathscr{O}_W)$, and W is said to be a *subvariety*. This simply combines the open and the closed case:

$$U \cap Z \underset{\text{closed subvar.}}{\subseteq} U \underset{\text{open subvar.}}{\subseteq} X.$$

Example 3.9. $k^2 \smallsetminus \{0\} = D(x) \cup D(y)$ is an open subvariety of $k^2 = \mathrm{Sp}(k[x, y])$ covered by two affine opens. But it is not itself affine.

Example 3.10. The union of the three coordinate planes minus the origin is a (non-affine) subvariety of k^3:

$$V(xyz) \smallsetminus \{(0,0,0)\} \subset k^3$$



Question. Why care for varieties when we are mostly interested in LAGs which are affine?

Answer. Quotients of LAGs are often not affine anymore.

Example 3.11. Consider $G = \mathrm{GL}_2$, $X = k^2 \smallsetminus \{(0,0)\}$, and the action

$$G \times X \xrightarrow{\mu} X, \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} := \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}.$$

Note that μ is a morphism of varieties because it comes by restriction from a polynomial map $G \times k^2 \to k^2$.

Fix the vector $x := (1\ 0)^t \in X$. We observe that

- X is single orbit, i.e. $G \cdot x = X$ (as set), and that
- the stabilizer

$$G_x = \{g \in G \mid gx = x\} = \underbrace{\mu^{-1}(x) \cap G \times \{x\}}_{\text{both closed}}$$

is closed subgroup of G, hence again a LAG. In this case:

$$G_x = \left\{ \begin{pmatrix} 1 & b \\ & d \end{pmatrix} \right\} \subseteq \mathrm{GL}_2.$$

So we see that

$$\underbrace{G/G_x \xrightarrow{\sim} X}_{\text{will be formalized later}} \qquad \text{is a non-affine variety!}$$

3.3. **Constructible subsets.** Reference: [1, §4.3, 4.4].

Question: Let $\varphi : Y \to X$ be a morphism of varieties. Which properties does $\varphi(Y)$ have?

Theorem 3.12. *The possible images $\varphi(Y)$ for varying Y, φ are precisely the* constructible subsets *of X.*

Definition 3.13. A subset $W \subseteq X$ is called *constructible* if it can be written as a finite union of locally closed subsets.

The constructible subsets are precisely those which can be obtained from open subsets and finitely many operations of the form taking union, intersection, or complement.

Example 3.14. A constructible set which is not locally closed would be

$$\big(V(xyz) \cap D(x)\big) \cup \{(0,0,0)\} \subset k^3.$$

Proof of Theorem 3.12. Clearly, if $W = W_1 \cup \cdots \cup W_r$ is finite union of locally closed subsets W_i, then each W_i is variety and

$$W = \mathrm{Im}\left(\coprod_{i=1}^{r} W_i \longrightarrow X \right)$$

is trivially the image of a morphism. The meaningful direction is the converse. So let $\varphi : Y \to X$ be a morphism, our goal being to show that $\varphi(Y)$ is constructible.

Step 1: Reduction to affine case We cover X by finitely many affine varieties, $X = U_1 \cup \ldots \cup U_r$, $U_i \cong \mathrm{Sp}(A_i)$. Then

$$\varphi(Y) = \bigcup_{i=1}^{r} \varphi(\varphi^{-1}(U_i))$$

and it is enough to show that every set on the right hand side is constructible. This allows to assume X affine from now on.

Similarly, if $Y = V_1 \cup \ldots \cup V_r$ is a finite covering by affines, then

$$\varphi(Y) = \bigcup_{i=1}^{r} \varphi(V_i),$$

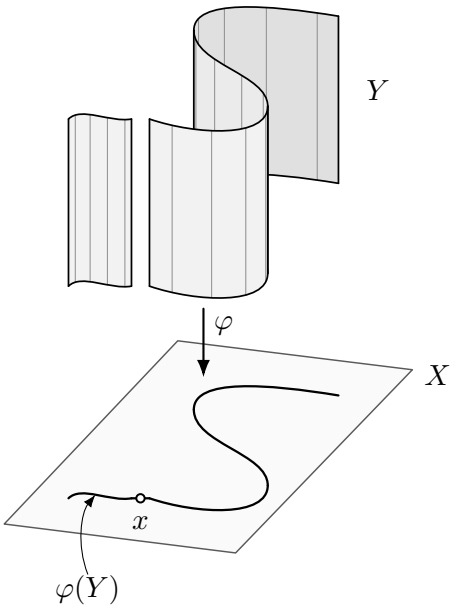
so we may also assume that Y is affine.

Step 2: Reduction to integral domains Write Y as a union of its irreducible components $Y_1, \ldots, Y_r$. Then

$$\varphi(Y) = \bigcup_{i=1}^{r} \varphi(Y_i),$$

so it suffices to consider the case when Y is irreducible. Then the image $\varphi(Y) \subseteq X$ is irreducible as well. It follows that the closure $\overline{\varphi(Y)}$ is irreducible as well. This closure is a closed subvariety of X, and it suffices to show that $\varphi(Y) \subseteq \overline{\varphi(Y)}$ is constructible.

Here is a sketch of the situation,



with $\overline{\varphi(Y)} = \varphi(Y) \cup \{x\}$

Replacing X by $\overline{\varphi(Y)}$, we have reduced to irreducible $Y = \mathrm{Sp}(B)$, $X = \mathrm{Sp}(A)$, and a morphism φ with dense image $\varphi(Y) \subseteq X$. Then A and B are integral domains by (2.1), and the pullback map $\varphi^* : A \to B$ is injective. Indeed, $\varphi^*(f) = 0$ means that φ factors through the vanishing set $V(f)$ which would be a *proper* closed subset if $f \neq 0$.

The main argument. Everything so far was formal and immediate, but now the actual work starts. Recall that $\varphi^* : A \hookrightarrow B$ is an injective map of integral domain finite type k-algebras.

Consider the field extension $\mathrm{Quot}(A) \subseteq \mathrm{Quot}(B)$. It is generated by B and has finite transcendence degree, say $r \geq 0$. Choose a transcendence basis $b_1, \ldots, b_r \in B$. Let $A[b_1, \ldots, b_r] \subseteq B$ be the A-subalgebra generated by $b_1, \ldots, b_r$. Since the b_i are algebraically independent, each of its elements is a unique polyonomial in the $b_1, \ldots, b_r$ with coefficients in A. In this way, we have factored φ as $p_X \circ \psi$:

$$Y \xrightarrow[\psi]{((b_1,\ldots,b_r),\varphi)} k^r \times X \xrightarrow{\quad p_X \quad} X$$

$$B \longleftarrow A[b_1, \ldots, b_r] \longleftarrow A$$

Step 3. We claim that $\psi(Y)$ contains an open subset of $k^r \times X$.

Proof. Since B is a finitely generated k-algebra, it is also a finitely generated $A[b_1, \dots, b_r]$-algebra. Let $\beta_1, \dots, \beta_m$ be a set of generators. Since $\mathrm{Quot}(B)/\mathrm{Quot}(A)(b_1, \dots, b_r)$ is finite, each β_i is the zero of some monic polynomial

$$T^{d_i} + c_{i,d_i-1}T^{d_i-1} + \cdots + c_{i,0},$$

with coefficients of the form $c_{i,j} = \frac{p_{i,j}}{q_{i,j}}$ for some $p_{i,j}, q_{i,j} \in A[b_1, \dots, b_r]$. Let

$$q := \prod_{i,j} q_{i,j}$$

be the product of all denominators. Then

- $B[q^{-1}]$ is still generated by $\beta_1, \dots, \beta_m$ as $A[b_1, \dots, b_r, q^{-1}]$-algebra.
- Each β_i is integral over $A[b_1, \dots, b_r, q^{-1}]$.

So $A[b_1, \dots, b_r, q^{-1}] \subseteq B[q^{-1}]$ is a finite extension of integral domains, and hence

$$\mathrm{Sp}(B[q^{-1}]) \longrightarrow\!\!\!\!\!\rightarrow \mathrm{Sp}(A[b_1, \dots, b_r, q^{-1}])$$

surjective by going up and going down. We see that $D(q) \subseteq k^r \times X$ lies in the image. □

Step 4. Consider an irreducible $X = \mathrm{Sp}(A)$ as before, the projection $p_X : k^r \times X \to X$, and a non-empty open subset $U \subseteq k^r \times X$. We claim that $p_X(U)$ contains an open subset of X.

Proof. Let $0 \neq q \in A[b_1, \dots, b_r]$ be a polynomial such that $D(q) \subseteq U$. Express it in standard form

$$q = \sum_{I \in \mathbb{Z}^r_{\geq 0}} a_I b^I.$$

Then $p_X(D(q)) = \bigcup_I D(a_I)$ because precisely for the points x of this union is $q(-, x) \in k[b_1, \dots, b_r]$ a non-zero polynomial and equivalently

$$D(q(-, x)) = k^r \times \{x\} \cap D(q)$$

non-empty. □

Step 5: End by induction. Combining Steps 3 and 4, we have shown that $\varphi(Y) \subseteq X$ contains an open subset. Since X is irreducible, $X \setminus U$ is of strictly smaller dimension than X. We end by induction. □

3.4. **Images are again LAGs.** Reference: [1, §7.4]

After these preparations, we can can wrap up the argument that images of homomorphisms are automatically closed.

Proposition 3.15. *Let $\varphi : G \to H$ be a homomorphism of LAGs. Then $\varphi(G) \subseteq H$ is a closed subgroup and, in particular, again naturally an affine variety.*

Proof. By Theorem 3.12, the image $\varphi(G) \subset H$ is constructible and a subgroup. We need a few lemmas.

Lemma 3.16. *Let $K \subset H$ be a set-theoretic subgroup of a LAG H. Then the closure $\overline{K} \subseteq H$ is a subgroup as well.*

Proof. By Corollary 3.3, the inverse map $i : H \longrightarrow H$ is a variety automorphism. In particular, it is a homeomorphism, so

$$\overline{K}^{-1} = \overline{K^{-1}} = \overline{K}.$$

This shows that $\overline{K}$ is stable under taking inverse. For $x \in K$, $x \cdot \overline{K} = \overline{xK} = \overline{K}$ since left multiplication by x is homeomorphism as well. Hence $K\overline{K} \subseteq \overline{K}$.

As a special case, this says that for $x \in \overline{K}$ we have $Kx \subseteq \overline{K}$. Since right multiplication by x is also a homeomorphism, we get that $\overline{Kx} = \overline{K}x \subseteq \overline{K}$. So after all $\overline{K} \cdot \overline{K} \subseteq \overline{K}$, showing that $\overline{K}$ is a group as claimed. □

Lemma 3.17. *Let X be a variety and $W \subseteq X$ a constructible subset. Then W contains a dense open subset of its closure $\overline{W}$.*

Proof. Left as exercise. □

Lemma 3.18. *Let $U, V \subseteq K$ be open subsets of LAG K with U being dense. Then $U \cdot V := \mathrm{Im}(U \times V \xrightarrow{m} K) = K$.*

Proof. For $x \in K$, by Corollary 3.3, the translated inverse $x \cdot i(V)$ is again an open subset. Since U is dense, the intersection $U \cap x \cdot i(V)$ is non-empty. This means there exist $u \in U$ and $v \in V$ such that $u = x \cdot v^{-1}$ as desired. □

Combining the various statements, we first obtain from Lemma 3.16 that $\overline{\varphi(G)} \subseteq H$ is closed. By Lemma 3.17, there exists an open dense subset $U \subseteq \overline{\varphi(G)}$ which is already contained in $\varphi(G)$. By Lemma 3.18, we have $U \cdot U = \overline{\varphi(G)}$. Since $\varphi(G)$ is a group to begin with, we get $\overline{\varphi(G)} \subseteq \varphi(G) \cdot \varphi(G) = \varphi(G)$ as desired. □

4. Actions and orbits (10/10/2026)

General problem. Let G be a LAG and $H \subset G$ a closed subgroup. How to endow the set of orbits G/H with a natural variety structure?

Definition 4.1. An *action of G* on a variety X is defined as a morphism

$$\mu : G \times X \longrightarrow X$$

that is also a set-theoretic action.

Definition 4.2. A *categorical quotient* for G acting on X is defined as a *universal* G-invariant morphism $\pi : X \to Q$. That is, for every G-invariant morphism $\psi : X \to Y$, there exists a unique morphism $\overline{\psi} : Q \to Y$ factoring ψ as

$$\begin{array}{ccc} X & \xrightarrow{\pi} & Q \\ & \searrow^{\psi} & \downarrow{\scriptstyle \overline{\psi}} \\ & & Y. \end{array}$$

In the above, *G-invariant* means that $\pi(gx) = \pi(x)$ for all $g \in G, x \in X$.

Remark 4.3. A categorical quotient (Q, π) is unique up to unique isomorphism because it is characterized by its universal property.

Definition 4.4. A categorical quotient $\pi : X \to Q$ is called a *geometric quotient* if π also provides a set-theoretic bijection

$$G\backslash X \xrightarrow{\sim} Q.$$

Exercise 4.5. The mutiplicative group acts on the affine line $\mathbb{A}^1$ by scaling, i.e.

$$\mathbb{G}_m \times \mathbb{A}^1 \longrightarrow \mathbb{A}^1, \quad \lambda \cdot x = \lambda x.$$

There are two orbits,

$$k^\times \quad \text{and} \quad \{0\}.$$

Show that the categorical quotient exists, but is equal to a point $\mathrm{Sp}(k) = \{\mathrm{pt}\}$. It is hence not a geometric orbit.

Precise problem. For a subgroup $H \subset G$, prove that the categorical quotient of G by H exists and is a geometric quotient. By Remark 4.3, this defines an unambiguous variety structure on G/H.

Special case. In the previous lecture, we showed that if $H = \ker(\varphi)$ is the kernel of a homomorphism of LAGs $\varphi : G \longrightarrow K$, then the image

$$K \overset{\text{closed}}{\supseteq} \varphi(G) = G/H$$

is closed and hence has a natural variety structure. Even then it is not obvious to see that $G \to \varphi(G)$ is a categorical quotient (e.g. independent of φ). Moreover, this construction only applies if H is a kernel and in particular at least a normal subgroup.

We proceed in three steps:

(1) Given $H \subset G$, we will construct a variety X with G-action such that H is the stabilizer of some $x \in X$.

(2) The set-quotient G/H can then be identified with the orbit $G \cdot x$. We prove that that $G \cdot x$ is locally closed to give it a variety structure.

(3) We use two more advanced results to show that this makes $G \cdot x$ into a categorical quotient.

4.1. **Orbits.** Throughout, Let LAG G acts on variety X, such that

$$X = \coprod_{G\cdot x \in G\backslash X} G \cdot x$$

is the disjoint union of orbits (as sets). Each $x \in X$ has a stabilizer $G_x \subseteq G$, and the isomorphism

$$G/G_x \xrightarrow{\sim} G \cdot x$$

as sets.

Lemma 4.6. *The stablizer $G_x \subset G$ of the G-action is a closed subgroup.*

Proof. Since $\{x\} \subset X$ is locally closed (closed in an affine neighborhood). Let

$$\mu_x : G \longrightarrow X, \qquad \mu_x(g) = g \cdot x.$$

Then the stablizer is the inverse image of μ_x

$$G_x = \mu_x^{-1}(\{x\}) \subset G_x,$$

hence it is locally closed. From previous lecture, if $H \subset G$ is a subgroup and constructible, then H is a closed subgroup. Apply to G_x we get the result. □

Proposition 4.7. *(a) Each G-orbit is locally closed in X and smooth with its subvariety structure.*

(b) The orbit $G\cdot x$ is equidimensional, i.e. all irreducible component have same dimension, and $\dim(G \cdot x) = \dim(G) - \dim(G_x)$.

(c) The closure of an orbit $\overline{G \cdot x}$ is G-stable. The boundary $\overline{G \cdot x} \setminus G \cdot x$ is a union of G-orbits of strictly smaller dimension than $G \cdot x$.

(d) The G-orbit $G \cdot x$ is the disjoint union (finite many) of its G°-orbits.

Proof. (a) By Theorem 3.12, $G \cdot x$ is constructible. Hence there exists an open $U \subset \overline{G \cdot x}$ contained in $G \cdot x$. (No need to choose dense, but possible.) That is, there is an open $\widetilde{U} \subseteq X$ such that

$$U = \widetilde{U} \cap \overline{G \cdot x}$$

is contained in $G \cdot x$.

Define

$$\widetilde{V} := \bigcup_{g \in G} g \cdot \widetilde{U}, \qquad V = \widetilde{V} \cap \overline{G \cdot x}.$$

Then $\widetilde{V}$ is open, so V is locally closed. But we also have

$$V = \bigcup_{g \in G} g \cdot U = G \cdot x.$$

Hence $G \cdot x$ is locally closed, in particular it is a subvariety.

Proposition 4.8. *Let $\varphi : Z \longrightarrow Y$ be a dominant morphism of varieties, here dominant means image dense i.e. $\overline{\varphi(Z)} = Y$. Then there exists a point $z \in Z$ in which φ is smooth by using* $\operatorname{char} k = 0$.

Recall. The morphism of varieties φ is smooth in $z \in Z$ if, for some (equivalently, all) choices of local coordinates of $z \in \operatorname{Sp}(B) \underset{\text{open}}{\subseteq} Z$ and $\varphi(z) \in \operatorname{Sp}(A) \underset{\text{open}}{\subseteq} Y$

$$B = A[T_1, \ldots, T_n]/(f_1, \ldots, f_m),$$

the rank of the Jacobi matrix in z,

$$J_z := \left(\frac{\partial f_i}{\partial T_j} \right)_{i,j} (z) \in M_n(k),$$

equals the codimension of the fibers,

$$\begin{aligned} \operatorname{codim}_z \big(Z_y \overset{T_i}{\subset} k^n \big) &:= n - \dim_z(Z_y) \\ &:= n - \text{Krull. dim} \big(B/\mathfrak{m}_y B \big)_{\mathfrak{m}_z}. \end{aligned}$$

Fact. For variety Y over an algebraically closed field k, Y/k is smooth if and only if Y is regular.

Special case. Let $Y = \{pt\} = \operatorname{Sp}(k)$, Z is smooth in z iff Z is regular in z meaning that the local ring $B_{\mathfrak{m}_z}$ is regular.

Back to the proof of Proposition 4.7.

(a) We have shown the orbit $G \cdot x \subseteq X$ is locally closed, hence it is a subvariety. By the Proposition 4.8, there exists a smooth (regular) point $y \in G \cdot x$. Since G acts on $G \cdot x$ transitively, every point of $G \cdot x$ is smooth. □ (a)

(b) The Proposition 4.8 also gives that

$$\mu_x : G \longrightarrow G \cdot x, \qquad g \longmapsto gx,$$

is smooth. Namely, there exists a smooth point $g \in G$ for μ_x. For every element $h \in G$ defines a commutative square

$$\begin{array}{ccc} G & \xrightarrow[\sim]{h\cdot -} & G \\ \downarrow & & \downarrow \\ G\cdot x & \xrightarrow[\sim]{h\cdot -} & G\cdot x \end{array}$$

So by translation, $\mu_x : G \to G \cdot x$ is smooth, hence every point is smooth.

In general, if $\varphi : Z \longrightarrow Y$ is smooth in z, with $y = \varphi(z)$,

$$\dim_z(Z) = \dim_z(Z_y) + \dim_y(Y).$$

In our setting, the fiber over $g \cdot x$ is the conjugate stabilizer gG_xg^{-1}, which is isomorphic to G_x, so get

$$\dim_g(G) = \dim_g(\underbrace{gG_xg^{-1}}_{\mu_x^{-1}(gx)}) + \dim_{gx}(G \cdot x).$$

□ (b)

(c) We see that each $g \cdot - : X \to X$ acts as a homeomorphism, it commutes with taking closure, so

$$g \cdot \overline{G \cdot x} = \overline{gG \cdot x} = \overline{G \cdot x} \quad \Longrightarrow \quad \overline{G \cdot x} \text{ is } G\text{-stable.}$$

The second part of the statement is automatic. In general, for a constructible subset $W \subset X$,

$$\dim(\overline{W}) = \dim(W) \qquad \text{and} \qquad \dim(\overline{W} \setminus W) < \dim(W).$$

Here, we define $\dim(W)$ as $\max \dim(W_i)$ if $W = \bigcup_i W_i$ with W_i locally closed. □ (c)

(d) We can regard orbit $G \cdot x$ as sets, which can be decomposed as

$$G \cdot x = \bigcup_{y \in G/G^\circ} G^\circ \cdot yx$$

into G°-orbits. The claim is that each $G^\circ \cdot yx \subseteq G \cdot x$ is open and closed for the subspace topology.

For the Closedness. If $z \in \overline{G^\circ \cdot yx} \setminus G^\circ \cdot yx$, then $\dim(G_z^\circ) > \dim(G_x^\circ)$ by (b) and (c) applied to G°. We get

$$\dim(G^\circ \cdot z) < \dim(G^\circ \cdot yx)$$

hence

$$(G_x)^\circ = (G^\circ)_x.$$

Applying to x and z, we get $\dim(G_z) > \dim(G_x)$, and hence $z \in \overline{G \cdot x} \setminus Gx$ by (b) and (c).

For the Openness. It is clearly by taking complement:

$$G^\circ \cdot yx = G \cdot x \setminus \bigcup_{G^\circ \cdot y'x \neq G^\circ \cdot yx} G^\circ \cdot y'x.$$

□

Corollary 4.9. *There exists at least one closed orbit.*

Proof. Begin with any orbit $G \cdot x$. If it is not closed, replace x by some $x' \in \overline{G \cdot x} \setminus G \cdot x$. Then $\dim(G \cdot x') < \dim(G \cdot x)$ by (c) in Proposition 4.7. Then we can iterate the argument to get the result. □

4.2. **Natural examples.**

(1) Consider the group action $\mathbb{G}_m \curvearrowright k^n$ by

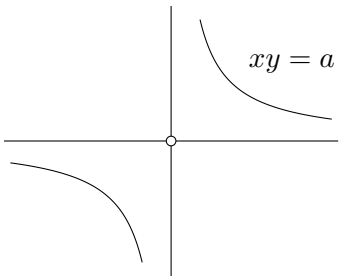
$$\lambda \cdot (x_1, \ldots, x_n) = (\lambda x_1, \ldots, \lambda x_n).$$

The orbits are $\{0\}$ (closed point) and $\ell \setminus \{0\}$ for $\ell = k \cdot v \in k^n$ a line. If $n \geq 2$, the latter are neither open nor closed.

(2) For the case $n = 2$, the group action $\mathbb{G}_m \curvearrowright k^2$ by

$$\lambda \cdot (x, y) = (\lambda x, \lambda^{-1} y).$$

The orbits: closed point $\{(0,0)\}$ and 1 dimensional hyperbolas $\{xy = a \mid a \in k^\times\}$, and non-closed 1 dimensional axis $\{x \neq 0, y = 0\}, \{x = 0, y \neq 0\}$.



(3) Let V be a finite-dimensional k-vector space, with $\mathrm{GL}(V)$ an algebraic group. The group action $\mathrm{GL}(V) \curvearrowright V$ has orbits: $\{0\}$ is closed, and $V \setminus \{0\}$ open.

(4) For $\dim(V) \geq 2$, the group action $\mathrm{GL}(V) \curvearrowright V^2$ is defined by

$$g(v, w) = (gv, gw).$$

gives the following orbits: The open orbit

$$\{(v, w) \mid v, w \text{ linearly independent}\},$$

which has dimension $2n$, the closed n-dimensional orbits for $a \in k \cup \{\infty\}$,

$$l_a^\times = \begin{cases} \{(v, av)\} & v \neq 0 \text{ if } a \in k \\ \{(0, w)\} & w \neq 0 \text{ if } a = \infty, \end{cases}$$

and the closed point

$$\{(0,0)\}.$$

In fact, see an example of $G \cdot x$ such that $\overline{G \cdot x} \setminus G \cdot x$ has infinitely many G-orbits.

For $G = \mathrm{GL}_2 \curvearrowright V = k^2$, the vectors

$$v := \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}, \quad w := \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \text{ are linearly independent}$$

if and only if $\det(v, w) = v_1 w_2 - v_2 w_1 \neq 0$. So we have a morphism $\det : V^2 \longrightarrow k$, equivariant in the sense

$$\det(gv, gw) = \det(g) \det(v, w).$$

The preimage $\det^{-1}(k^\times)$ is an open orbit. The kernel $\det^{-1}(0)$ is a quadric

$$v_1 w_2 - v_2 w_1 = 0.$$

which is a union of lines ℓ_a, for $a \in k \cup \{\infty\}$, it is also a union of orbits $\ell_a \setminus \{0\}$ and $\{0\}$.

(5) Given that $H \in \mathrm{Sym}_n(k)$, with $\det(H) \neq 0$. The quadratic form is defined as

$$Q(v) := \frac{1}{2} v^t H v.$$

The special orthogonal group $G := \mathrm{SO}(H)$ acts on k^n, and clearly the map

$$Q : k^n \longrightarrow k$$

is invariant. In fact,

$$G \cdot v = G \cdot w \quad \Longrightarrow \quad Q(v) = Q(w).$$

Conversely, assume v, w nonzero with $Q(v) = Q(w) =: q$.

• If $q \neq 0$ (anisotropic): Then the vector space k^n decompose

$$k^n = \langle v \rangle \oplus \langle v \rangle^\perp = \langle w \rangle \oplus \langle w \rangle^\perp.$$

Both $\langle v \rangle^\perp$, $\langle w \rangle^\perp$ are non-degenerate quadratic spaces, since k is algebraically closed:

$$k^n \cong (k^{n-1}, \mathbb{1}_{n-1}) \cong \langle w \rangle^\perp.$$

So there exists an isometry

$$\varphi : \langle v \rangle^\perp \xrightarrow{\sim} \langle w \rangle^\perp,$$

with $\det(\varphi) = 1$. Then $g : k^n \longrightarrow k^n$, defined by

$$g(v) = w, \qquad g|_{\langle v \rangle} = \varphi.$$

Namely, for $\lambda \in k$, $v \in \langle w \rangle^\perp$

$$\begin{aligned}(g(\lambda v + v^\perp), g(\lambda v + v^\perp)) &= \lambda^2 (gv, gv) + 2\lambda (gv, gv^\perp) + (gv^\perp, gv^\perp) \\ &= \lambda^2 (w, w) + 2\lambda (w, \varphi(v)^\perp) + (\varphi(v^\perp), \varphi(v^\perp)) \\ &= \lambda^2 (v, v) + 0 + (v^\perp, v^\perp)\end{aligned}$$

This implies $g \in \mathrm{SO}(H)$. Hence $G \cdot v = G \cdot w$.

• If $q = 0$ (isotropic): using the similar argument. Since $\det(H) \neq 0$, there exists an element $e' \in k^n$ such that

$$(v, e') \neq 0,$$

rescaling this, we can assume

$$(v, e') = 1.$$

Since the quadratic form Q is non-degenerate, then

$$(e' + \lambda v, e' + \lambda v) = (e', e') + 2\lambda.$$

Hence

$$e := e' - \frac{(e', e')}{2} v$$

is isotropic, $(e, e) = 0$, and it still satisfies $(v, e) = 1$. Hence

$$\langle v, e \rangle \cong \left(k^2, \begin{pmatrix} & 1 \\ 1 & \end{pmatrix} \right)$$

is a hyperbolic plane.

Since Q is non-degeneracy, we can take orthogonal complement,

$$k^n = \langle v, e_v \rangle \oplus \langle v, e_v \rangle^\perp = \langle w, e_w \rangle \oplus \langle w, e_w \rangle^\perp.$$

as before, and define g componentwise using the similar argument. Then $G \cdot v = G \cdot w$.

Now $Q : k^n \to k$ is categorical quotient $\mathrm{SO}(H)\backslash k^n$. Take $f \in k[k^n]$ to be $\mathrm{SO}(H)$-invaraint polynomial, then

$$f(-) = r(Q(-))$$

for some polynomial $r \in k[T]$. The orbits:

$$Q^{-1}(q) = \begin{cases} \text{single closed orbit if } q \neq 0 \\ \{0\} \sqcup \{ \text{ isotropic vectors } \} \text{ if } q = 0 \end{cases}$$

Remark 4.10. Here the condition k algebraically closed is not necessary, here we just used for simplicity.

(6) Consider the group action of $\mathrm{GL}_n \curvearrowright \mathrm{M}_n(k)$ by conjugation,

$$g \cdot x := gxg^{-1}.$$

First, for $n = 2$. There exists a morphism

$$\mathrm{M}_2(k) \longrightarrow k^2, \qquad x \longmapsto \big(\mathrm{tr}(x), \det(x)\big).$$

We know the characteristic polynomial of matrix x,

$$\mathrm{char}(x, T) = T^2 - \mathrm{tr}(x)T + \det(x)$$

is separable if and only if the discriminant

$$\Delta(x) := \mathrm{tr}(x)^2 - 4\det(x) \neq 0.$$

The discriminant Δ is a morphism $k^2 \to k$,

$$\Delta(t, d) := \Delta(T^2 + tT + d) = t^2 - 4d.$$

For (t, d) with $\Delta(t, d) \neq 0$, have a closed 2-dimensional orbit

$$X_{t,d} := \{x \in \mathrm{M}_2(k) \mid \mathrm{tr}(x) = -t,\ \det(x) = d\}.$$

Here, we know that x, y with separable characteristic polynomial are conjugate if and only if their characteristic polynomials are equal.

Now consider the non-separable case $4d = t^2$.

$$X_{t,d} = \underbrace{\mathrm{GL}_2 \cdot \begin{pmatrix} t/2 & 1 \\ & t/2 \end{pmatrix}}_{\text{2-dimensional orbit}} \sqcup \underbrace{\left\{ \begin{pmatrix} t/2 & \\ & t/2 \end{pmatrix} \right\}}_{\substack{\text{closed orbit,} \\ \text{0-dimensional}}}.$$

For the general case of n, there exists a map

$$\mathrm{M}_n(k) \longrightarrow k^n, \qquad x \longmapsto (a_{n-1}, \ldots, a_0)$$

where $a_{n-1}, \ldots, a_0$ are coefficients of characteristic polynomial of x, i.e.

$$\mathrm{char}(x, T) = T^n + a_{n-1}T^{n-1} + \cdots + a_0.$$

We claim there exists a discriminant polynomial

$$\Delta : k^n \longrightarrow k$$

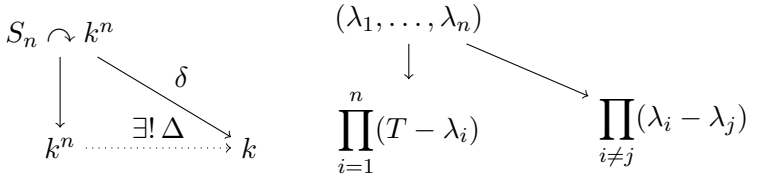
such that the discriminant

$$\Delta(a_{n-1}, \ldots, a_0) \neq 0$$

if and only if the characteristic polynomial

$$T^n + a_{n-1}T^{n-1} + \cdots + a_0 \text{ separable.}$$

The following diagram describes the situation, let S_n be the symmetric group acting on k^n,



Note that $\lambda_1, \ldots, \lambda_n$ are zeros of a degree n polynomial, but it is only unique up to permutation. The $\delta(\lambda_1, \ldots, \lambda_n) \neq 0$ is obviously equivalent to $\prod_{i=1}^{n}(T-\lambda_i)$ separable. Hence δ is also S_n-symmetric. By *Theorem on elementary symmetric functions*, there exists a unique polynomial $\Delta = \Delta(a_{n-1}, \ldots, a_0)$ such that

$$\delta(\lambda_1, \ldots, \lambda_n) = \Delta\left(\text{coefficients of } \prod_{i=1}^{n}(T-\lambda_i)\right).$$

Then the discriminant $\Delta(\mathrm{char}(x)) \neq 0$ if and only if the characteristic polynomial is separable for all separable $P \in k^n$, with the unique closed orbit $X_P \subseteq M_n(k)$. In general, X_P is a finite union of orbits described by possible Jordan Normal Forms for characteristic polynomial P.

5. GL_n, PROJECTIVE, QUOTIENTS (14/10/2026)

Example 5.1. *Claim: There exists no non-trivial homomorphism from* GL_n *to* $\mathbb{G}_a$. i.e. $\mathrm{Hom}(\mathrm{GL}_n, \mathbb{G}_a) = 0$.

The proof of claim contains the following steps:

(1): First show that $\mathrm{Hom}(\mathbb{G}_m, \mathbb{G}_a) = 0$,

$$\mathrm{Hom}(\mathbb{G}_m, \mathbb{G}_a) = \{a(t) \in k[t, t^{-1}] : a(xy) = a(x) + a(y)\}$$

Thus it is clearly 0.

(2): There is a diagonal torus $T = \mathbb{G}_m^n \subset \mathrm{GL}_n$, Step (1) shows that any morphism $\varphi : \mathrm{GL}_n \to \mathbb{G}_a$ satisfies $\varphi|_T = 0$. Moreover, $\mathbb{G}_a$ abelian, so $\varphi(xyx^{-1}) = \varphi(y)$ morphism, for all $x, y \in \mathrm{GL}_n$. So y diagonalizable implies $\varphi(y) = 0$.

There is an observation that is diagonalizable elements are dense, with following inclusions:

$$\{\text{diagonalizable matrix with different eigenvalues}\} \subset \{\text{ diagonalizable matrix}\} \subset \mathrm{GL}_n.$$

The diagonalizable matrix with different eigenvalues is equivalnet to the open $D(\Delta)$ where Δ is discriminant, which imples $\varphi = 0$.

This method can also works to show $\mathrm{Hom}(\mathrm{SO}_n, \mathbb{G}_a)$ and $\mathrm{Hom}(\mathrm{Sp}_{2m}, \mathbb{G}_a)$ vanish (HWS).

5.1. **Projective varieties.**

Definition 5.2. A projective n-space $\mathbb{P}^n$ is defined as the lines in k^{n+1}, i.e.

$$\mathbb{P}^n \cong k^\times \backslash (k^{n+1} \setminus \{0\}),$$

with point in form $[x_0, \ldots, x_n]$. For given k-vector space V, the projective space $\mathbb{P}(V)$ is defined as all lines in V.

In topology perspects: For any homogeneous polynomial $f \in k[X_0, \ldots, X_n]$, the vanishing locus of f is

$$V(f) := \{[x_0 : x_1 : \cdots : x_n] \mid f(x_0, \ldots, x_n) = 0\}$$

and the open $D(f) := V(f)^c$, is the complement of vanishing locus. All these opens $\{D(f)\}$ form basis for the Zariski topology on projective n-space $\mathbb{P}^n$.

Theorem 5.3 (Hilbert)**.** *There is one to one correspondence between radcal homogeneous ideal of $k[X_0, \ldots, X_n]$ to the closed subsets of projective space $\mathbb{P}^n$,*

$$\begin{aligned} \{\textit{Radical homogeneous ideal } I \subseteq k[X_0, \ldots, X_n]\} &\overset{1:1}{\longleftrightarrow} \{\textit{closed } Z \subseteq \mathbb{P}^n\} \\ I &\longmapsto V(I) \\ I(Z) &\longmapsto Z \end{aligned}$$

Variety structure. Let f be non-zero homogeneous polynomial

$$k[D(f)] = k[X_0, \ldots, X_n, f^{-1}]_{\deg 0}$$

Example 5.4. The function $\frac{y^2+z^2}{x^2}$ defines a subvariety of $\mathbb{P}^3$ which take values $\frac{10}{4}$ at the point $[2 : 1 : 3]$ is an affine line $k \cdot \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$.

Question. How to describe an affine neighborhood of $l \in \mathbb{P}(V)$ for $\dim V = n + 1$?

Answer. Choose the complement of l, i.e. $V = l \oplus W$, and take the basis for l denote as e_0 and basis for W denote $e_1, \ldots, e_n$. Line snot contained in W are of the form

$$k \cdot \left(e_0 + \sum_{i=1}^{n} t_i e_i \right)$$

for unique $t_1, \ldots, t_n \in k$, thus this defines an affine chart

$$\mathrm{Sp}(k[T_1, \ldots, T_n]) \xrightarrow{\sim} \{l' \not\subset W\} \overset{\text{open}}{\subset} \mathbb{P}(V).$$

The RHS above is isomorphic to $\mathbb{P}(V) \setminus \mathbb{P}(W)$ and contains line l.

Example 5.5. Sometimes a morphism is directly given in projective coordinates:

$$\mathbb{P}^1 \longrightarrow V(XY - Z^2) \subset \mathbb{P}^2, \qquad [a : b] \longmapsto [a^2 : b^2 : ab].$$

We can check the morphism property in charts, e.g. on open subset $D(a)$, we get the map

$$\begin{aligned} \mathbb{A}^1_k \cong D(a) &\longrightarrow D(x) \cong \mathbb{A}^2_k \\ [1 : b] &\longmapsto [1 : b^2 : b] \end{aligned}$$

Here is an observation for the situation, is the closed immersion

$$k \xrightarrow{\sim} V(y - z^2) \subset k^2$$

similar on other charts implies the closed immersion

$$\mathbb{P}^1 \xrightarrow{\sim} V(XY - Z^2).$$

The generalization of this situation is the *Veronese embedding*:

$$\begin{aligned} \mathbb{P}^n &\hookrightarrow \mathbb{P}^{\binom{n+d}{d}-1} \\ [x_0 : \cdots : x_n] &\longmapsto [x^I], \end{aligned}$$

where $x^I := x_0^{a_0} \cdots x_n^{a_n}$, $I = (a_0, \ldots, a_n)$, with $|I| = d$.

Definition 5.6. A *projective variety* is defined to be isomorphic to a closed subvariet of some projective space $\mathbb{P}^n$. A *quasi-projective variety* is defined to be isomorphic to a locally closed subvariety of some projective space $\mathbb{P}^n$.

Example 5.7. The affine variety is equivalent to the closed subvariety in some $\mathbb{A}^n = \mathbb{P}^n \setminus \mathbb{P}^{n-1}$, hence it is quasi-projective.

5.2. **Products.** Recall $X = \mathrm{Sp}(A), Y = \mathrm{Sp}(B), Z = \mathrm{Sp}(Z)$, the product $X \times Y = \mathrm{Sp}(A \otimes B)$, and the fiber product

$$Y \times_X Z = \mathrm{Sp}((B \otimes_A C)_{red}).$$

Now we can extend to general varieties:

Let $X = \cup U_i$, $Y = \cup V_j$, where U_i, V_j are open affine. The product $X \times Y$ is equal to the the product of sets with the variety structure $\cup_{i,j} U_i \times V_j$. Consider the X, Y are S-schemes, with structure morphism f, g,

$$\begin{array}{ccccc} X & & & & Y \\ & \searrow^{f} & & \swarrow_{g} & \\ & & S & & \end{array}$$

We can discuss about the covers on this diagram:

(1) S can be covered by open affine i.e.

$$S = \cup_k S_k.$$

(2) The preimage admits a cover i.e.

$$f^{-1}(S_k) = \bigcup_i U_{ki}, \quad g^{-1}(S_k) = \bigcup_j V_{kj}.$$

(3) We declare that

$$\bigcup_k \bigcup_{i,j} U_{ki} \times_{S_k} V_{kj}$$

as the open affine covering of $X \times_S Y$.

Definition 5.8. A *Segre embedding* is

$$\begin{aligned} \mathbb{P}^n \times \mathbb{P}^m &\hookrightarrow \mathbb{P}^{nm+m+n} \\ ([x_i], [y_j]) &\longmapsto [x_i y_j]_{i,j} \end{aligned}$$

Example 5.9. The product of projective varieties is projective. Let X, Y be projective varieties of $\mathbb{P}^m$ and $\mathbb{P}^n$ for some m, n,

$$X \times Y \underset{\text{closed}}{\subset} \mathbb{P}^m \times \mathbb{P}^n \hookrightarrow \mathbb{P}^{mn+m+n},$$

where the last arrow is Segre embedding.

5.3. **Grassmanians.** Reference: [1, §1.8].

Let V be an n-dimensional k-vector space, we define the *Grassmanian*

$$\mathrm{Gr}_d(V) = \{W \subset V \text{ d-dimensional subspace of } V\}.$$

This becomes a variety for the following affine open covering. Given $W \in \mathrm{Gr}_d(V)$, we can choose the complement of W in V, i.e. $V = W \oplus U$. We declare that the following set

$$\{W' \in \mathrm{Gr}_d(V) \mid W' \cap U = \{0\}\}$$

is an open set.

We can choose bases $e_1, \ldots, e_d \in W$, and $e_{d+1}, \ldots, e_n \in U$. Then every W' is spanned by vectors of form

$$e_i + \sum_{i=d+1}^{n} t_{ij} \cdot e_j$$

for unique coefficient t_{ij}, $1 \leq i \leq d$, $d+1 \leq j \leq n$.

Use these t_{ij} to identify

$$k^{d(n-d)} = \mathrm{Sp}(k[T_{1,d+1}, \ldots, T_{d,n}]) \xrightarrow{\sim} \{W' \mid W' \cap U = \{0\}\}$$

and check that this defines variety structure on $\mathrm{Gr}_d(V)$.

Proposition 5.10. *The Grassmanian $\mathrm{Gr}_d(V)$ is projective. That is, there exists an embedding into some $\mathbb{P}^N$.*

Definition 5.11. Let V be a vector space, and let m be an integer, the *m-th exterior power* is a vector space $\wedge^m V$ plus the universal alternating map

$$V^m \longrightarrow \bigwedge^m V$$

satisfies the following condition that for every alternating map $V^m \to W$, there exists a unique k-linear map $\wedge^m V \to W$ such that the following diagram commutes

$$\begin{array}{ccc} V^m & \longrightarrow & \wedge^m V \\ \downarrow & \swarrow_{\exists!} & \\ W & & \end{array}$$

Here the alternating morphism means

•multilinear and $(\ldots, v, \ldots, v, \ldots) = 0$ for every vector $v \in V$, outside of the characteristic 2 field, if and only if $(\ldots, v, \ldots, w, \ldots) = -(\ldots, w, \ldots, v, \ldots)$.

• Notation of universal mapping : $(v_1, \ldots, v_m) \mapsto v_1 \wedge \cdots \wedge v_m$.

• If $e_1, \ldots, e_n$ are basis of V, then $e_{i_1} \wedge \cdots \wedge e_{i_m}$, for $i_1 < \cdots < i_m$ form a basis of $\wedge^m V$, hence

$$\dim \bigwedge^m V = \binom{\dim V}{m}.$$

• Functoriality: for any $\varphi : V \to W$, there is a commutative diagram

$$\begin{array}{ccc} V^m & \xrightarrow{\varphi^m} & W^m \\ \downarrow & & \downarrow \\ \wedge^m V & \xrightarrow{\exists!\, \wedge^m \varphi} & \wedge^m W \end{array}$$

The morphism $\wedge^m \varphi$, sends $v_1 \wedge \cdots \wedge v_m$ to $\varphi(v_1) \wedge \cdots \wedge \varphi(v_m)$.

• In paticular, the Grassmanian is under the construction

$$\mathrm{Gr}_d(V) \longrightarrow \mathrm{Gr}_{\binom{d}{m}}(\bigwedge^m V)$$
$$W \subset V \longmapsto \bigwedge^m W \subset \bigwedge^m V.$$

Thus is a closed immersion, which we verify for $m = d$.

We now construct the *Plücker embedding* $\mathrm{Gr}_d(V) \hookrightarrow \mathbb{P}^{\binom{n}{d}-1}$.

Proof of Proposition 5.10. Consider the morphism

$$\begin{aligned} \mathrm{Gr}_d(V) &\longrightarrow \mathbb{P}(\bigwedge^d V) \\ W &\longmapsto \bigwedge^d W. \end{aligned}$$

First claim: This morphism is injective. Take two d-dimensional subspaces W and W', we choose bases of them, where the coordinates of W is $e_1, \dots, e_m, e_{m+1}, \dots, e_d$ and W' with coordinates $e_1, \dots, e_m, e'_{m+1}, \dots, e'_d$. So the intersection $W \cap W'$ has base $e_1, \dots, e_m$. Then

$$e_1 \wedge \cdots \wedge e_m \wedge e_{m+1} \wedge \cdots \wedge e_d$$

and

$$e_1 \wedge \cdots \wedge e_m \wedge e'_{m+1} \wedge \cdots \wedge e'_d$$

are two different vectors of lines of $\bigwedge^d V$, and hence generate different lines.

Next claim: The morphism is a closed immersion.

We have to check in a local chart around $W \subset V$, with coordinates $e_1, \dots, e_d \in W$ and $e_{d+1}, \dots, e_n \in V$.

$$\begin{aligned} k^{d(n-d)} &\xrightarrow{\varphi} k^{\binom{n}{d}} \\ \mathrm{span}\left\{ e_i + \sum_{j=d+1}^{n} t_{ij} e_j \right\} &\longmapsto \mathrm{span}\left\{ (e_1 + \sum_{j=d+1}^{n} t_{1j} e_j) \wedge \cdots \wedge (e_d + \sum_{j=d+1}^{n} t_{dj} e_j) \right\} \end{aligned}$$

The RHS is equal to the following sum

$$e_1 \wedge \cdots \wedge e_d + \sum_{i=1}^{d} \sum_{j=d+1}^{n} t_{ij} e_1 \wedge \cdots \wedge e_{i-1} \wedge \hat{e_j} \wedge e_{i+1} \wedge \cdots \wedge e_n + \text{higher degree in } e_{d+1}, \dots, e_n.$$

So in this ordering, the first few coordinates of φ are

$$\varphi(t_{1,d+1}, \dots, t_{d,n}, \underbrace{\cdots\cdots}_{\text{polynomials in } t_{ij}})$$

The morphism on rings

$$k[T_{ij}] \xleftarrow{\varphi^*} k[X_{i_1,\dots,i_d} \mid 1 \le i_1 \le \cdots \le i_d \le n, (i_1, \dots, i_d) \neq (1, 2, \dots, d)]$$

with $\varphi^*(X_{1,\dots,\hat{i},\dots,d,j}) = (-1)^{j+i+1} T_{ij}$, here $X_{1 \le i_1 < \cdots < i_d \le n}$ with $i_d > d$ corresponds the $e_{i_1} \wedge \cdots \wedge e_d$. Hence it is surjective. The sign comes from

$$e_1 \wedge \cdots \wedge e_{i-1} \wedge e_j \wedge e_{i+1} \wedge \cdots \wedge e_d = (-1)^{i+j+1} e_1 \wedge \cdots \wedge \hat{e_i} \wedge \cdots \wedge e_d \wedge e_j.$$

On rings, this is surjective, hence it is a closed immersion on Spectrum. □

5.4. **Constructing quotient as variety.** Reference: [1, §11]

The matrix action $\mathrm{GL}(V) \curvearrowright \mathrm{Gr}_d(V)$ given by

$$g \cdot W = g(W)$$

is algebraic and transitive. So we have constructed new quotients, such as

$$\mathrm{GL}_n / \left\{ \begin{pmatrix} \star_d & \star \\ 0 & \star_{n-d} \end{pmatrix} \right\} \xrightarrow{\sim} \mathrm{Gr}_d(k^n)$$
$$g \longmapsto g \cdot \langle e_1, \ldots, e_d \rangle$$

The projective setting also contains our previous affine examples:

$$\mathrm{GL}_n = \left\{ \begin{pmatrix} 1 & \\ & \star_n \end{pmatrix} \right\} \subset \mathrm{GL}_{n+1}$$

preserves and acts naturally on open $D(x_0) \cong k^n \subset \mathbb{P}^n$.

Proposition 5.12. *For every pair $H \subset G$, there exists a projective variety X with G-action and a point $x \in X$ such that $H = G_x$, i.e. the stablizer of some point x.*

Proof. Since $H \subset G$ is a closed subvariety, so $k[H] = k[G]/I$ for some radical ideal I. □

Lemma 5.13. *The closed subvariety H has the form*

$$\{x \in G \mid \rho_x(I) = I\} \qquad \rho_x(f)(y) = f(yx)$$

here ρ_x is the right translation operator acting on the ring $k[G]$.

Proof. This is immediate because

$$H = \{x \in G \mid H \cdot x = H\} = \{x \in G \mid \rho_x \text{preserves ideal defining } H \subset G\}$$

Let $W \subset k[G]$ be finite dimensional, stable under ρ_x for all $x \in G$, and it contains generator set of I. Then $M := W \cap I$ is stable under all $\rho_x, x \in H$. Conversely if $x \in G$ satisfies $\rho_x(M) = M$, then

$$\rho_x(I) = \rho_x(M \cdot k[G]) = \rho_x(M)\rho_x(k[G]) = M \cdot k[G] = I,$$

so $x \in H$ by previous lemma, then H can be written in form

$$\{x \in G \mid \rho_x \text{ stablizes } M \subset W\},$$

so $H = G_M$ for $M \in \mathrm{Gr}_d(W)$ with $\dim(M) = d$. □

Remark 5.14. If it is densed, can pass to $\wedge^d W$ and $\wedge^d M \in \mathbb{P}(\wedge^d W)$ so that induces an action $G \curvearrowright \mathbb{P}^N$ for some N and $x \in \mathbb{P}^N$ satisfied that $H = G_x$.

5.5. **Quotient property.** Reference [1, §12]

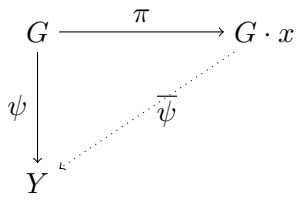
We have already obtained the following statement: For all $H \subset G$ on the action of G on the projective vareity X and $x \in X$ such that $H = G_x$. In particular, the geometric quotient exists as

$$G/H \xrightarrow{\sim} G \cdot x.$$

We see that the stablizer $G \cdot x \subseteq X$ is locally closed, hence it is a *quasi-projective variety.*

Theorem 5.15. *$G \xrightarrow{\pi} G \cdot x$ is a categorical quotient $G \to G/G_x$.*

Proof. We need to show that for every caetgroical quotient $\pi : G \to G \cdot x$ and for any H-invaraint map $\psi : G \to Y$ there exists a unique morphism $\overline{\psi}$ such that the following diagram commutes:



The uniqueness of the morphism is clear: For $x \in \mathrm{Gr}_d(W)$, the action is on set $\overline{\psi}(gx) = \psi(g)$.

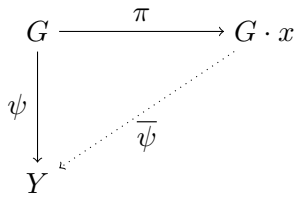
The $\overline{\psi}$ is continuous: We need a non-trivial commutative algebra statement (*Stack project* [2] and [3]): Flat morphism between finitely generated algebras has open image, that is, if $A \xrightarrow{\varphi} B$ is flat and $U \subset \mathrm{Sp}(B)$ is open, then $\mathrm{Sp}(\varphi)(U) \subset \mathrm{Sp}(A)$ is open.

□

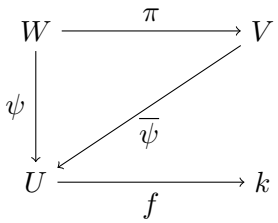
Application. Let $U \subset Y$ be open. Then $\overline{\psi}^{-1}(U) = \pi(\psi^{-1}(U))$ because π is surjective . But this is open by continuity of ψ and the fact just cited for the π, we know π is smooth, which implies that π is flat. Since we consider the varieties, so finitely presentation is automatically.

To prove $\overline{\psi}$ is a variety morphism, we can assume $G = G^\circ$ is connected.

Here is the situation



implies the following diagram



Here, $W = \psi^{-1}(U)$ and $V = \overline{\psi}^{-1}(U)$, and $f : U \to k$ is regular. Now we need to show that the composition $f \circ \overline{\psi} : V \to k$ is regular. We know that $f \circ \psi = f \circ \overline{\psi} \circ \pi$ is regular and π is flat and surjective. This turns out to be sufficient.

Claim: for every open $V \subset G \cdot x$ with a morphism $h : V \to k$ such that the composition

$$h \circ \pi : \pi^{-1}(V) \to k$$

is regular. Then h is regular.

Proof of the claim. We can cover V by open affines, since regularity is local property, so we can assume V is affine. Then $W = \pi^{-1}(V)$ is affine as well. For thus we need to go through some more definitions

Definition 5.16. We say a variety X is separated that means the image of diagonal morphism

$$\Delta : X \longrightarrow X \times X$$

is closed.

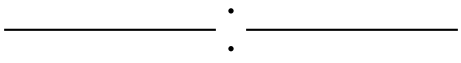
Exercise 5.17. Here are some exercises for the diagonal morphism:

(a) The variety X is affine, then X is separated.

(b) Let $Z \subseteq X$ be a subvariety, that is, Z is a locally closed subset of X. If X is separated, then Z is separated.

(c) The projective space $\mathbb{P}^n$ is separated, by Exercise (b) above, any quasi=projective variety is separated.

(d) The variety of a line with double origins $k^\times \cup \{0_1, 0_2\}$



is not separated.

Remark 5.18. The diagonal morphism Δ is always locally closed.

• If $\varphi : Y \to X$ is a morphism and X is separated, then the graph of φ, $\Gamma_\varphi \subset Y \times X$, is closed because the Cartesian diagram

$$\begin{array}{ccc} \Gamma_\varphi & \longrightarrow & Y \times X \\ \downarrow & & \downarrow (f, \mathrm{id}) \\ \Delta(X) & \longrightarrow & X \times X \end{array}$$

• Let X be separated, with two open affine $U, V \subset X$, then $U \cap V$ is affine since the following Cartesian diagram

$$\begin{array}{ccc} U \cap V & \longrightarrow & U \times V \\ \downarrow & & \downarrow \\ \Delta & \longrightarrow & X \times X \end{array}$$

• If X is separated, Y is affine, and $\varphi : Y \to X$ is a morphism of varieties. Let $U \subset X$ be an affine cover, then $\varphi^{-1}(U) \subset Y$ is also affine, with the following Cartesian,

$$\begin{array}{ccc} \varphi^{-1}(U) & \longrightarrow & Y \times U \\ \downarrow & & \downarrow \\ \Delta & \longrightarrow & X \times X \end{array}$$

Back to our claim, since G is affine and the orbit $G \cdot x$ is separated, then W is affine. Due to the morphism π is flat and surjective so for every regular function $h \in B$, h is constant on the fiber of π.

Next we claim : $h \in A$ We use an important tool which is called *faithfully flat descent.*

Theorem 5.19. *The k-algebra A can be expressed as the descent*

$$\{h \in B \mid h \otimes 1 = 1 \otimes h \in B \otimes_A B\}.$$

In our setting, we have the following equalizer form of affine varieties

$$\mathrm{Sp}\left((B \otimes_A B)_{\mathrm{red}}\right) \overset{p_1}{\underset{p_2}{\rightrightarrows}} \mathrm{Sp}(B) \longrightarrow \mathrm{Sp}(A)$$

with $h \circ p_1 = h \circ p_2$. So we need to see that $B \otimes_A B$ is reduced.

In our case, let A, B are integral domains, with the following inclusion

$$B \otimes_A B \hookrightarrow B_K \otimes_K B_K$$

where $K = \mathrm{Quot}(A)$, and $B_K := B[(A \setminus \{0\})^{-1}]$.

By Noetherian normalization, we know

$$K \subset K[x_1, \ldots, x_m] \subset B_K$$

Then the tensor product gives

$$B_K[y_1, \ldots, y_m] \otimes_{K[x_1, \ldots, x_m, y_1, \ldots, y_m]} B_K[x_1, \ldots, x_m] \hookrightarrow L(y_1, \ldots, y_m) \otimes_{K(x_1, \ldots, y_m)} L(x_1, \ldots, x_m)$$

here $L = \mathrm{Quot}(B)$.

Now since we consider k is characteristic 0, the tensor product of finite field extensions is reduced.

Eventually, for the group action on separated variety $G \curvearrowright X$ with $x \in X$. Then the orbit $G \cdot x \subset X$ is locally clsoed, hence it is a variety and the morphism $G \to G \cdot x$ is geometric quotient G/G_x.

Moreover $G \cdot x$ has quotient topology, with $V \subset G \cdot x$ is affine so that the preimage $\pi^{-1}(V)$ is affine and

$$k[V] = k[\pi^{-1}(V)]^{G_x}$$

is a translation invariant functions.

In particular, if $G \cdot x$ is affine, then $G \cdot x = \mathrm{Sp}(k[V]^{G_x})$.

Remark 5.20. The property of separatedness may be unnecessary

Proposition 5.21. *If $H \subset G$ is normal, then G/H is again LAG.*

Proof. Use categorical quotient for

$$G \times G \longrightarrow G/H \times G/H.$$

There is a commutative diagram

$$\begin{array}{ccc} G \times G & & \\ \downarrow & \searrow & \\ G/H \times G/H & \overset{\exists!}{\dashrightarrow} & G/H \end{array}$$

□

This is a morphism inducing group structure on the quotient G/H so we get a LAG-structure on quotient as desired. □

References

[1] Humphreys, James E.; *Linear algebraic groups*, Graduate Texts in Mathematics **21**, Springer-Verlag New York-Heidelberg,1975.
[2] The Stacks Project Authors; *Stacks Project*, Tag 01UA, https://stacks.math.columbia.edu/tag/01UA.
[3] The Stacks Project Authors; *Stacks Project*, Tag 00I1, https://stacks.math.columbia.edu/tag/00I1.